

SYSTEMS STUDIES AND OPTIMIZATION OF THE ARIES-CS POWER PLANT

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Received April 16, 2007

Accepted for Publication December 10, 2007

A stellarator systems/optimization code is used to optimize the ARIES-CS fusion power plant parameters for minimum cost of electricity subject to a large number of physics, engineering, and in-vessel component constraints for a compact stellarator configuration. Different physics models, reactor component models, and costing algorithms are used to test sensitivities to models and assumptions. The most important factors determining the size of the fusion power core are the allowable neutron and radiative power fluxes to the wall, the distance needed between the edge of the plasma and the nonplanar magnetic field coils for the intervening components, and an adequate tritium breeding ratio. The magnetic field and coil parameters are determined from both plasma performance and constraints on the Nb₃Sn superconductor. The same costing approach and algorithms used in previous ARIES studies are used with updated material costs. The result is a compact stellarator reactor with a major

radius close to that of tokamaks. A one-dimensional power balance code is used to study the path to ignition and the effect of different plasma and confinement assumptions on plasma performance for the reference plasma and coil configuration. A number of variations are studied that affect the size and cost of the fusion power core: maximum field at the coils, component cost penalties, a different blanket and shield approach, alternative plasma and coil configurations, etc. Comparisons are made with some earlier ARIES power plant studies. A number of issues for the development of compact quasi-axisymmetric stellarators are identified.

KEYWORDS: compact stellarator, fusion power plant, parameter optimization

Note: The figures in this paper are in color only in the electronic version.

I. INTRODUCTION

The confining poloidal magnetic field in currentless stellarators is created by currents in external nonplanar magnetic field coils, resulting in a nonaxisymmetric plasma with a noncircular cross section that varies within a toroidal field period. The plasma is characterized by an average major radius $\langle R_{axis} \rangle$ and an average plasma radius $\langle a \rangle$. Separate toroidal field coils are not required and poloidal (“vertical”) field coils may be included for flexibility and position control. Compact stellarators¹ are

low-aspect-ratio stellarator-tokamak hybrids with the potential for an attractive, fully ignited reactor. Because the confining poloidal magnetic field is created by currents in external windings, aided by a small plasma bootstrap current, compact stellarators are inherently steady-state devices without the need for the large plasma current of the tokamak and spherical torus approaches. Compact stellarators may combine the best features of high-current tokamaks (moderate plasma aspect ratios, good confinement, and high volume-average plasma beta $\langle \beta \rangle$) with those of large-aspect-ratio currentless stellarators (steady-state high-plasma-density operation without external current drive or disruptions, stability against

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external kinks and vertical displacement events without a close conducting wall or active feedback systems, and low recirculating power in a fusion power plant).

Although stellarators have significant potential advantages as fusion power plants, earlier stellarator power plant studies led to large reactor sizes. The German HSR reactor study² had $\langle R_{axis} \rangle = 22$ m in a five-field-period ($M = 5$) embodiment and $\langle R_{axis} \rangle = 18$ m in a more recent $M = 4$ version. The $M = 4$ ARIES Stellarator Power Plant Study (SPPS) reactor³ had a smaller $\langle R_{axis} \rangle$ ($= 14$ m) due to its larger plasma-coil spacing compared to other stellarators. It was a first step toward a smaller-size reactor and was calculated to be cost competitive⁴ with the $R = 6$ m ARIES-IV and $R = 5.5$ m ARIES-RS tokamak reactors. A more compact stellarator reactor should retain the cost savings associated with the low recirculating power of the SPPS reactor but with smaller size and higher wall power density (and hence lower cost of electricity) by taking advantage of newly developed plasma and coil configurations and an improved blanket and shield concept.

II. PARAMETER OPTIMIZATION APPROACH

The ARIES-CS study uses the stellarator systems/optimization code⁵ to minimize the cost of electricity (CoE) or $\langle \beta \rangle$ for a chosen plasma and coil geometry by using a nonlinear constrained optimizer and iterating on a number of plasma and power plant variables. This re-

quires integration of the configuration properties, plasma properties and performance, reactor component constraints, and costing. Figure 1 illustrates the optimization procedure.

II.A. Plasma and Coil Geometry

The first step in the optimization procedure is choosing a particular plasma and coil geometry. Figure 2 shows a top view of the plasma and nonplanar modular coils for the reference configuration chosen for the ARIES-CS study, a 24-coil, three-field-period NCSX-based⁶ plasma configuration with coils modified to allow more space for the blankets and shield (the “ARE” configuration⁷). As with other stellarators, the plasma is nonaxisymmetric with a noncircular cross section, which is needed to create the required rotational transform ($\iota = 1/q$, where q is the tokamak safety factor) without a large toroidal plasma current. To create the necessary poloidal field from external coils, the coils must be nonplanar with large toroidal excursions and bends with small radii of curvature. Additional poloidal field coils are not needed for the steady-state equilibrium configuration. The compact stellarator variant studied here is quasi-toroidally symmetric⁸; the strength of the magnetic field $|\mathbf{B}|$ varies little in the toroidal coordinate direction in magnetic field line coordinates, analogous to the axisymmetry of $|\mathbf{B}|$ in tokamaks, spherical tori, and reverse field pinches. Other compact stellarator configurations are possible (quasi-poloidally symmetric⁹) as fusion power plants¹⁰ but were not examined in this study.

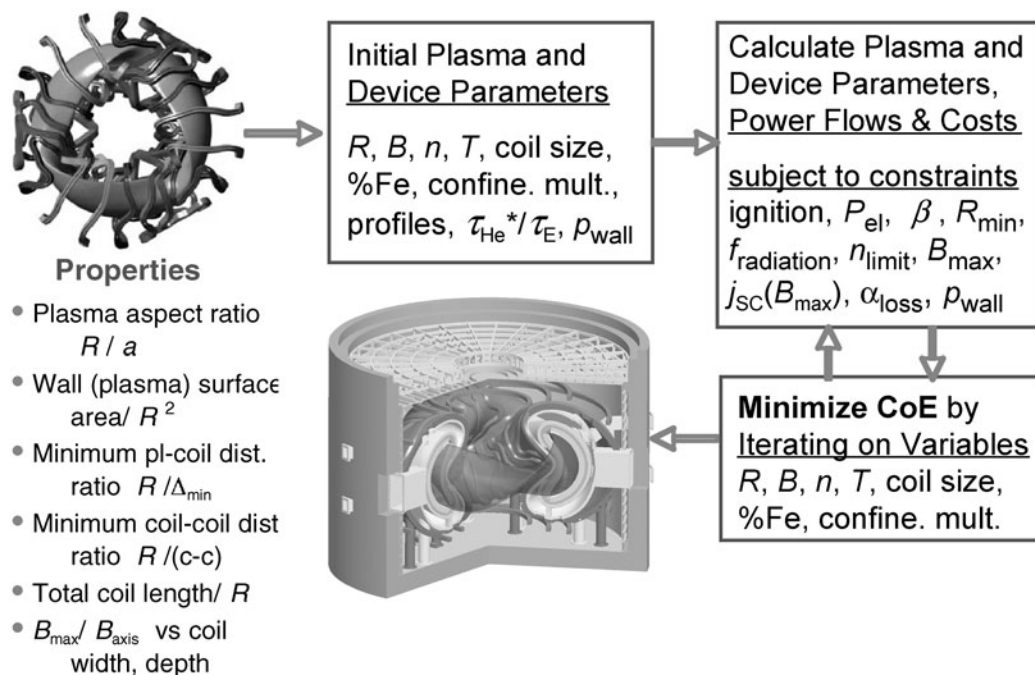


Fig. 1. Flowchart of the optimization procedure used in the ARIES-CS study.



Fig. 2. The ARE compact stellarator plasma and coil configuration.

The plasma and coil geometry of a stellarator is characterized by a set of dimensionless parameters. The plasma geometry enters through the shape of the last closed flux surface and the coil geometry enters through the shape of the modular coils. These dimensionless ratios are as follows:

1. the plasma aspect ratio $A_p = \langle R_{axis} \rangle / \langle a \rangle$
2. $A_{surf} / \langle R_{axis} \rangle^2$, where A_{surf} is the area of the last closed flux surface
3. $A_\Delta = \langle R_{axis} \rangle / \Delta_{min}$, where Δ_{min} is the minimum distance between the edge of the plasma and the current center of the modular coil winding pack
4. $A_{c-c,min} = \langle R_{axis} \rangle / d_{coil-coil}$, where $d_{coil-coil}$ is the minimum distance between the centers of the two closest modular coils
5. $L_{coil} / \langle R_{axis} \rangle$, where L_{coil} is the total length of all the coils in the modular coil set
6. $B_{max} / \langle B_{axis} \rangle$, a function of d and k , where B_{max} is the maximum field on the modular coil winding pack, d^2 is the cross-sectional area of the winding pack, and k is the toroidal elongation (toroidal width/radial depth) of the winding pack.

Table I gives these ratios for the reference ARIES-CS plasma and coil configuration (ARE) and two other plasma and coil configurations that were studied in less detail⁷: SNS, a higher-plasma-aspect-ratio, three-field-period configuration with low magnetic shear, and MHH2, a lower-plasma-aspect-ratio, two-field-period configuration with simpler coils that are nearly equidistant from the plasma along the coil winding. The other dimensionless parameters in the table are defined in the text where they are used.

II.B. Evaluation of Plasma and Device Parameters

Next, the plasma performance and power plant parameters are evaluated for a set of initial variables— $\langle R_{axis} \rangle$, $\langle B_{axis} \rangle$, the volume-averaged density $\langle n \rangle$, the density-averaged ion and electron temperature $\langle T \rangle$, the coil winding pack dimensions (toroidal width and radial depth), the plasma impurity fraction and type, the multiplier H for the stellarator confinement scaling used, the average neutron wall loading $\langle p_{n,wall} \rangle$, etc.—and some fixed parameters—the density and temperature profiles, the transport model, the ratio of the helium ash particle confinement time to the global energy confinement time τ_{He}^* / τ_E , the blanket and shield geometry and composition, the superconducting coil model $j(B_{max})$, and costing algorithms. The “fixed” parameters can be varied to test the sensitivity to these parameter choices, as discussed in Secs. IX and X.

Finally, a set of optimization variables— $\langle R_{axis} \rangle$, $\langle B_{axis} \rangle$, $\langle n \rangle$, $\langle T \rangle$, coil radial depth, and impurity fraction—are varied over a wide range to minimize the CoE (or $\langle \beta \rangle$) subject to a number of often conflicting constraints: equalities for the ignition condition (no power input) and the desired $P_{electric}$, and inequalities for the allowed limits on $\langle n \rangle$, H , $\langle \beta \rangle$, the average current density j in the superconducting modular coils versus B_{max} , the allowed plasma-coil and coil-coil distances, the tritium breed ratio (TBR), $p_{n,wall}$, the fraction of power radiated, the α -particle energy loss fraction, etc.

The nonlinear constrained optimization can be sensitive to the choice of initial parameter values because an initial starting point may not be connected to a minimum CoE point that satisfies all the constraints. In addition, the optimization can be overconstrained if the constraints are conflicting, but a converged solution that meets all the constraints is usually possible for a range of initial parameter values.

III. CONSTRAINTS ON THE MINIMUM SIZE FOR THE ARIES-CS FUSION POWER CORE

The most important factor in minimizing the CoE is the cost of the main power core components (first wall and divertor, blanket, shield, manifolds, vacuum vessel, coil winding pack, and coil support structure). For approximately fixed thicknesses of all but the coil pack, the volumes (hence costs) of these components are proportional to the wall area ($\propto \langle R_{axis} \rangle^2$). In addition, the volume of the modular coil winding pack is proportional to $L_{coil} I_{coil} / j_{coil} \propto \langle R_{axis} \rangle^{1.2}$. Minimizing the CoE amounts to minimizing the value of $\langle R_{axis} \rangle$, subject to a number of constraints. A secondary factor in minimizing the CoE for a given blanket and shield configuration is the dependence of the cost of the superconducting winding pack (discussed in Sec. IV) and the coil support structure ($\propto \langle B_{axis} \rangle^2$) on the strength of the magnetic field.

TABLE I
Plasma and Modular Coil Properties for ARIES-CS Reactors

Configuration	MHH2	ARE	SNS
Number of toroidal field periods	2	3	3
Plasma aspect ratio $\langle R_{axis} \rangle / \langle a \rangle$	2.66	4.55	6.00
Plasma current, $I_p / [\langle R_{axis} \rangle \langle B_{axis} \rangle]$ (MA/m·T)	0.201	0.072	0.043
Normalized plasma volume / $\langle R_{axis} \rangle^3$	2.793	0.950	0.548
Normalized plasma surface area / $\langle R_{axis} \rangle^2$	18.547	11.780	8.994
$\iota(s=0)$; $s \approx [\langle r \rangle / \langle a \rangle]^2$	0.33	0.42	0.52
$\iota(s=0.67)$	0.48	0.64	0.57
$\iota(s=1.0)$	0.58	0.66	0.55
Maximum effective helical ripple ε_{eff} (%)	~0.8	~0.6	~0.4
α -particle energy loss (%), model calculation	~5	~10	~7
Number of different coil types	4	3	3
Number of coils/period	8	6	6
Total number of coils	16	18	18
Coil aspect ratio: $\langle R_{axis} \rangle / \Delta_{min}$ (coil-plasma)	5.55	5.938	6.02
Coil separation ratio: $\langle R_{axis} \rangle / \Delta_{min}$ (coil-coil)	10.33	10.03	9.8
Normalized maximum coil current: $I / R \cdot B$ (MA/m·T)	0.316	0.306	0.278
Normalized coil lengths: $L / \langle R_{axis} \rangle$			
Coil 1	6.14	4.77	3.61
Coil 2	5.83	5.08	3.55
Coil 3	5.52	5.14	3.43
Coil 4	5.25	—	—
Normalized winding surface area / $\langle R_{axis} \rangle^2$	40.1	27.0	21.0
Normalized maximum field on coils $B_{max} / \langle B_{axis} \rangle$			
0.2 m $\times$ 0.2 m	4.27	4.02	3.64
0.3 m $\times$ 0.3 m	2.69	2.63	2.54
0.4 m $\times$ 0.4 m	1.94	2.10	2.18
0.5 m $\times$ 0.5 m	1.72	1.85	2.00
0.6 m $\times$ 0.6 m	1.60	1.70	1.89
0.8 m $\times$ 0.8 m	1.46	1.59	1.77

An important constraint on the minimum allowable value for $\langle R_{axis} \rangle$ is the need for adequate space D between the plasma edge and the coil center for the plasma scrape-off layer, first wall, blanket, shield, vacuum vessel, plasma-facing coil structure, and assembly gaps and half the radial depth of the coil winding pack. For a given plasma and coil configuration, the geometric ratio $A_\Delta = \langle R_{axis} \rangle / \Delta_{min}$ is a constant, where Δ_{min} is the value of the plasma-coil center spacing where the coils are closest to the plasma. This gives a value for $\langle R_{axis} \rangle_{min} = A_\Delta D$.

It is not possible to arbitrarily increase the value for Δ_{min} in order to reduce $\langle R_{axis} \rangle_{min}$. Increasing the plasma-coil spacing requires more convoluted coils with a higher spatial harmonic content to create the desired magnetic field spectrum at the plasma surface because the higher spatial harmonics of the magnetic field from the coils decay faster with distance from the coils. This is illustrated in Fig. 3, where the plasma aspect ratio A_p was held constant and the plasma-coil center aspect ratio A_Δ was varied from 5.68 to 6.82 for an NCSX-type plasma.⁷ The other curves in Fig. 3 are for the SNS ($A_\Delta = 6.02$) and MHH2 ($A_\Delta = 5.5$) configurations discussed in

Sec. X.F. A smaller A_Δ results in a larger value for $B_{max} / \langle B_{axis} \rangle$ and hence a lower $\langle B_{axis} \rangle$ for a given B_{max} value. Since $P_{fusion} \propto \langle B_{axis} \rangle^4 \langle R_{axis} \rangle^3$ for a fixed value of $\langle \beta \rangle$, $\langle R_{axis} \rangle$ must increase to compensate for the decrease in P_{fusion} due to a lower $\langle B_{axis} \rangle$.

The reference plasma and coil configuration permits a tapered blanket and shield that allows reducing $\langle R_{axis} \rangle$ because the plasma is close to the coils over only a small part of the wall area. Figure 4 shows a cutaway view of the fusion power core at the 0-deg bean-shaped plasma cross section. The minimum value of Δ occurs twice in a half field period, at 11 and 33 deg. There is a large space between the plasma surface and the coils everywhere except near the indented region. For $\langle R_{axis} \rangle > 10.1$ m, there is adequate space for a full blanket and shield even in this region. However, as $\langle R_{axis} \rangle$ decreases, the area available for a full blanket and shield decreases and a tapered blanket and shield is necessary to obtain a TBR > 1.1 . Figure 5 illustrates the two-region blanket and shield geometry¹¹ developed for the reference ARIES-CS case. The distance needed for a 5-cm scrape-off layer, blanket, shield, manifolds, assembly gaps, vacuum vessel wall,

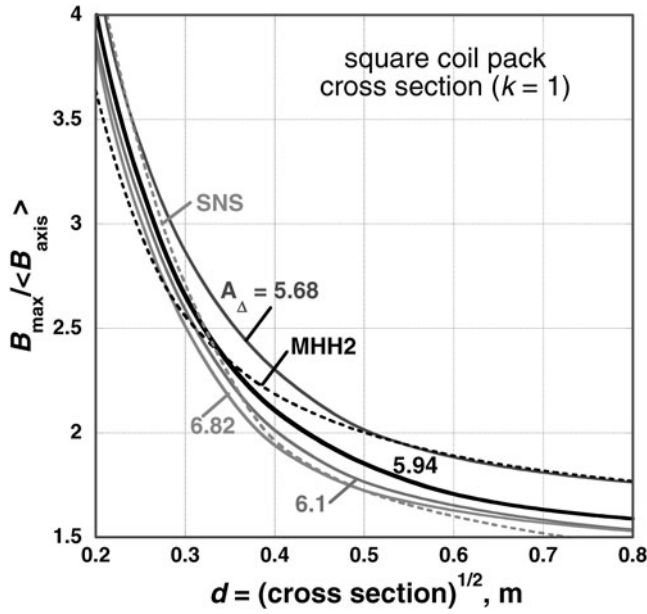


Fig. 3. Variation of $B_{\max}/\langle B_{\text{axis}} \rangle$ with A_{Δ} for an NCSX-type plasma and the SNS and MHH2 configurations.

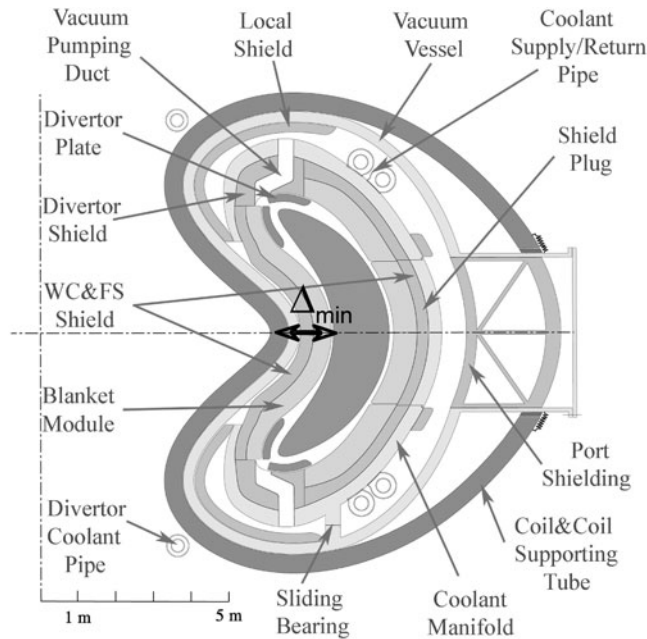


Fig. 4. Cutaway view of the fusion power core at the bean-shaped plasma cross section.

plasma-facing coil structure, and 9.7-cm half-radial depth of the coil pack is 1.305 m; since $A_{\Delta} = 5.938$, this gives $\langle R_{\text{axis}} \rangle_{\min} = 7.75$ m. For $\langle R_{\text{axis}} \rangle = 7.75$ m, the non-uniform tapered blanket covers 24% of the first-wall area. The extent of the tapered blanket region depends on

$\langle R_{\text{axis}} \rangle$. The contours in Fig. 6 show the wall region in the toroidal-angle poloidal-angle plane inside which a full blanket and shield *cannot* be accommodated, so a tapered blanket and shield is required there. This area increases with decreasing $\langle R_{\text{axis}} \rangle$ and the global TBR decreases. For a TBR > 1.1, $\langle R_{\text{axis}} \rangle$ must be > 7.5 m for 90%-enriched ${}^6\text{Li}$ in the blanket and shield.¹¹

Another constraint is the acceptable maximum power flux at the wall ($\propto 1/\langle R_{\text{axis}} \rangle^2$) for radiation (≤ 1 MW/m²), which is a wall-cooling limit, and for neutrons (≤ 5.4 MW/m²), which is a blanket lifetime issue. This gives a value for $\langle R_{\text{axis}} \rangle_{\min} \propto p_{n,\text{wall,max}}^{1/2}$. The $p_{n,\text{wall,max}} \leq 5.4$ MW/m² constraint requires $\langle R_{\text{axis}} \rangle \geq 7.75$ m for the 1-GW(electric) reference (ARE) power plant. There are no independent constraints on the peak power on the divertor, which is a serious design consideration, as discussed in Secs. V.D and IX.B. The peak power on the divertor depends on the ratio of the peak power to the average power on the divertor plates and the total power to the divertor, which depends on the power in particles impacting on the divertor (in lost alpha particles and power in the scrape-off layer that is not radiated) plus the radiated power that is intercepted by the divertor. The total power to the divertor is minimized through minimizing the fraction of lost alpha-particle power and the choice of the fraction of the power radiated from the plasma and radiated in the scrape-off layer. This is discussed in more detail in Sec. V.D.

The net effect of the various constraints on $\langle R_{\text{axis}} \rangle$ is that only values of $\langle R_{\text{axis}} \rangle \geq 7.75$ m are permitted, which allows us to reduce the enrichment of the ${}^6\text{Li}$ in the blanket and shield from 90% to 70% and still obtain a TBR > 1.1. Since the CoE increases with $\langle R_{\text{axis}} \rangle$, $\langle R_{\text{axis}} \rangle = 7.75$ m was chosen for the reference ARIES-CS case.

IV. DETERMINATION OF THE WINDING PACK PARAMETERS

The minimum value for $\langle R_{\text{axis}} \rangle$ results from extending the toroidal width w_t of the winding pack as much as possible, which minimizes the radial depth d_r of the winding pack and thereby $\langle R_{\text{axis}} \rangle$. The toroidal width of the winding pack is constrained by the normalized minimum coil-center to coil-center distance $A_{c-c,\min} = \langle R_{\text{axis}} \rangle / d_{\text{coil-coil}}$, a geometrical constant for a given coil geometry. From Table I, $A_{c-c,\min}$ is 10.03 for the ARE reference coil configuration. For $\langle R_{\text{axis}} \rangle = 7.75$ m, the value for $d_{\text{coil-coil}}$ is 0.773 m. Allowing a 3-cm space between coil winding packs where they are closest to allow for the structure into which the coils are wound gives $w_t = 0.743$ m. From Fig. 5, the minimum distance needed for the scrape-off layer, first wall, blanket, shield, assembly gap, vacuum vessel wall, and plasma-facing coil structure, which occurs at the $\Delta_{\min}$ location, is 1.208 m. For $\langle R_{\text{axis}} \rangle = 7.75$ m and the value of $A_{\Delta} = \langle R_{\text{axis}} \rangle / \Delta_{\min}$ in

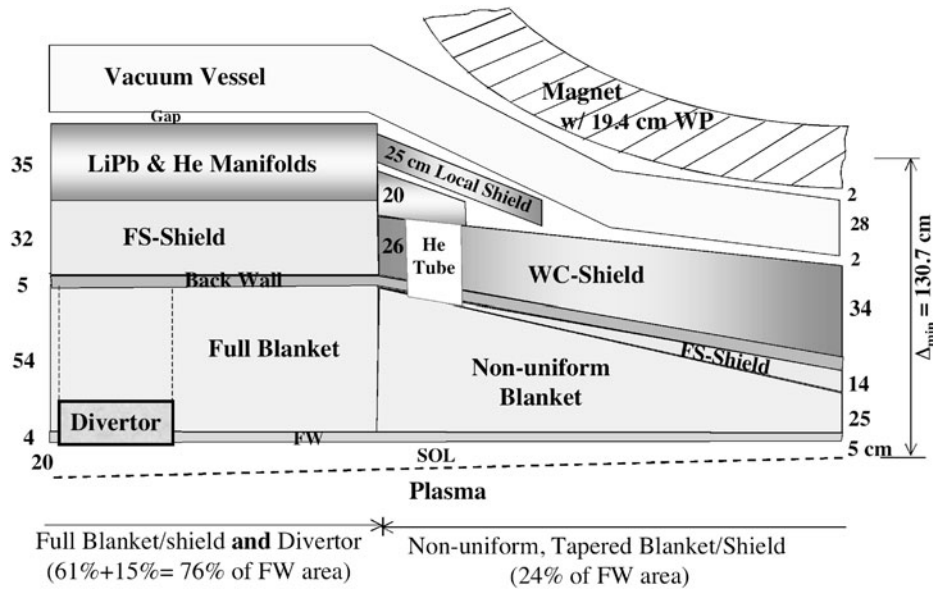


Fig. 5. Structure and nominal thicknesses for the LiPb/FS tapered blanket and shield. Here FS is ferritic steel. The first-wall end-of-life fluence limit is $15.7 \text{ MW} \cdot \text{yr}/\text{m}^2$, the overall TBR is 1.1, and the overall energy multiplication is 1.16.

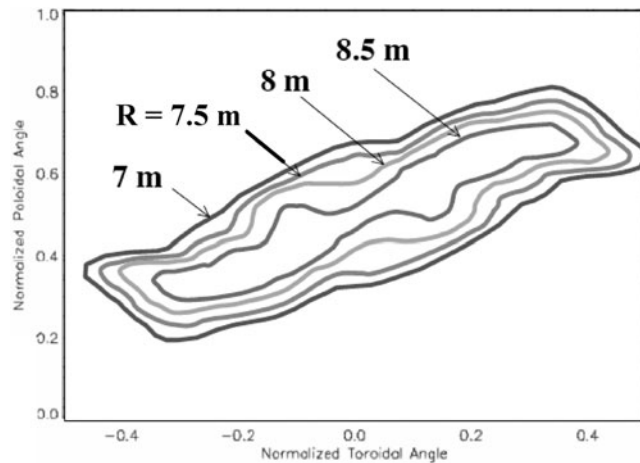


Fig. 6. Contour plot for tapered blanket areas.

Table I, the available distance between the plasma edge and the center of the coils is 1.305 m, which allows a maximum distance $d_r/2 = 9.7 \text{ cm}$, the value for the reference case. The resulting coil elongation is then $k = w_t/d_r = 3.83$. The values for d_r and k then determine $\langle B_{axis} \rangle / B_{max}$ by multiplying the $B_{max} / \langle B_{axis} \rangle$ obtained from Fig. 3 for a square coil pack ($k = 1$) with the value of $B_{max} / B_{max}(k = 1)$ versus k shown in Fig. 7.

The curves of $B_{max} / \langle B_{axis} \rangle$ in Figs. 3 and 7 are obtained from a filamentary coil model with 400 line segments and a correction for a finite-size square cross section coil pack. A three-dimensional (3-D) finite element analysis ANSYS model of the coils using 180 000 elements¹²

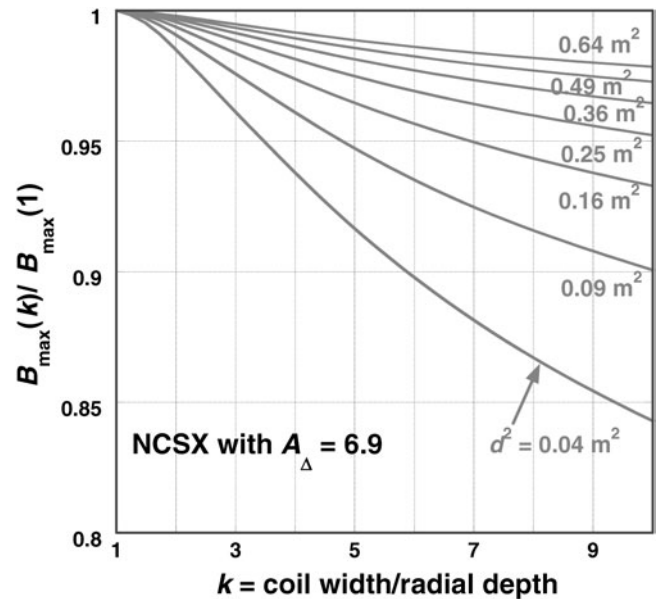
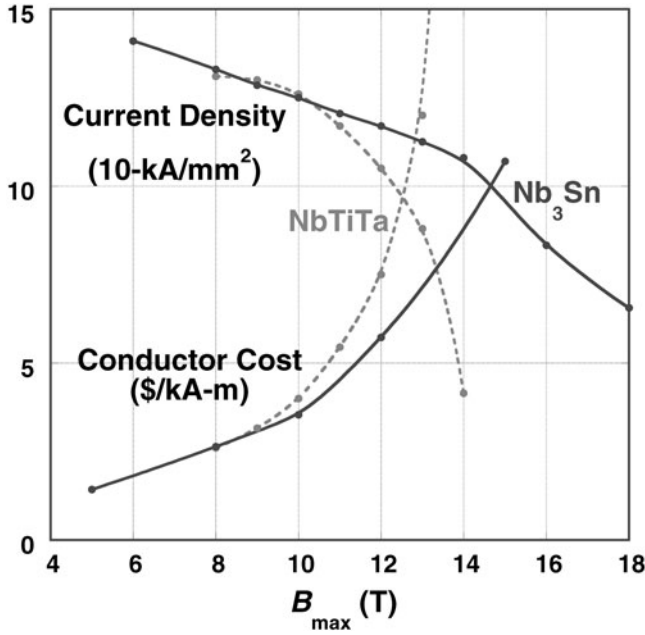


Fig. 7. Variation of $B_{max} / \langle B_{axis} \rangle$ with k , normalized to its value for $k = 1$, for one of the NCSX configurations.

gives a longer length (by a factor 1.135) and a higher value for $B_{max} / \langle B_{axis} \rangle$ than the values in Table I because of the reduced minimum bend radius at the surface of the coil. The value of $B_{max} / \langle B_{axis} \rangle$ in the optimization calculations is multiplied by a factor 1.25 to match the more accurate B_{max} calculation for the real 3-D coil geometry. The sensitivity to the B_{max} correction is discussed in Sec. X.A.


 Fig. 8. Dependence of j_{coil} and cost/kA·m on B_{max} .

The maximum allowed B_{max} is determined by satisfying two constraints on the average current density j_{coil} in the winding pack. The maximum j_{coil} is related to B_{max} by the $j_{coil} \leq j_{SC}(B_{max})$ constraint on j_{coil} for a Nb₃Sn superconducting coil shown in Fig. 8. The other superconducting coil design limits (heat deposition, radiation damage, etc.) are also satisfied locally, as discussed in detail in Refs. 11 and 12. Nb₃Sn is chosen for the winding pack conductor¹² to allow higher values of j_{coil} and B_{max} . The value of j_{coil} is also related to B_{max} through Ampere's law,

$$j_{coil} = [\langle B_{axis} \rangle / B_{max}] 2\pi \langle R_{axis} \rangle B_{max} / (\mu_0 N_{coils} d_r w_t) . \quad (1)$$

The intersection of the $j_{coil} = j_{SC}(B_{max})$ curve in Fig. 8 and that calculated from Eq. (1) gives $j_{coil} = 94$ MA/m² and $B_{max} = 15.1$ T, which is below the $B_{max} = 16$ T constraint. Knowing B_{max} then gives $\langle B_{axis} \rangle = 5.7$ T from the $B_{max}/\langle B_{axis} \rangle$ curves in Figs. 3 and 7.

V. PLASMA MODELS AND CONSTRAINTS USED IN CALCULATING PLASMA PERFORMANCE

The main power plant parameters (P_{fusion} , $\langle R_{axis} \rangle$, and $\langle B_{axis} \rangle$) place constraints on the plasma parameters. If the values for $\langle R_{axis} \rangle$ and $\langle B_{axis} \rangle$ are known from the considerations in the previous sections, then the value of $\langle \beta \rangle$ is also set because P_{fusion} is roughly proportional to $\langle \beta \rangle^2 \langle B_{axis} \rangle^4 \langle R_{axis} \rangle^3$ (this approximation is discussed later).

This also fixes τ_E because $\tau_E = W_{plasma} / P_{heating} \propto \langle n \rangle \langle T \rangle / P_{fusion} \propto \langle \beta \rangle \langle B_{axis} \rangle^2 / P_{fusion}$. A solution is not always possible that satisfies all the plasma and device constraints because the constraints can be conflicting for some assumptions, resulting in an overconstrained problem. This section discusses the various plasma constraints; the sensitivity of the ARIES-CS parameters to varying these constraints is discussed in Sec. IX.

The plasma performance depends on a number of assumptions: the scaling of the energy confinement time τ_E , the radial profiles and edge values for the plasma density and temperature, the plasma density and beta limits, the helium ash density, the scaling of the fraction of alpha-particle power that is lost, the fraction of alpha-particle power that is radiated, the impurity density profiles and values, and the fraction of power that is radiated in the scrape-off layer and in the vicinity of the divertor.

V.A. Energy Confinement Scaling

The plasma energy confinement is characterized by the ISS-95 stellarator confinement scaling¹³

$$\tau_E^{ISS-95} = 0.26 P_{heating}^{-0.59} \tilde{n}_e^{0.51} \langle B_{axis} \rangle^{0.83} \langle R_{axis} \rangle^{0.65} \langle a \rangle^{2.21} \iota_{2/3}^{0.4} , \quad (2)$$

with τ_E in s, $P_{heating}$ in MW, the line-average density $\tilde{n}_e$ in 10²⁰ m⁻³, $\langle B_{axis} \rangle$ in T, $\langle R_{axis} \rangle$ and $\langle a \rangle$ in m, and where $\iota_{2/3}$ is the rotational transform at $\langle r \rangle = 2/3 \langle a \rangle$. Figure 9 shows the experimental energy confinement times versus the empirical fit to these data for the stellarators in the 1995 international stellarator database and the ELMy H-mode database. Stellarators and tokamaks have similar plasma performance for similar device parameters (magnetic field, plasma volume, and heating power).

A confinement improvement factor H-ISS95 = τ_E / τ_E^{ISS-95} is determined in the ARIES-CS optimization from the global power balance. Data from the Wendelstein 7-AS (W7-AS) and the Large Helical Device (LHD) lie above the ISS-95 scaling (values for H-ISS95 up to 2.5 have been achieved). It is thought that lower values of the effective helical ripple (ϵ_{eff}) may play a role in the improved confinement. There is further evidence for this from analysis of the 2004 international stellarator database,¹⁴ which suggests that higher values of confinement improvement correlate with lower values of ϵ_{eff} ($\sim \epsilon_{eff}^{-0.4}$, as shown in Fig. 10). This correlation suggests that large H-ISS factors should be possible for the very-low- ϵ_{eff} quasi-symmetric compact stellarators.

V.B. Density and Temperature Profiles, Edge Values, and Density and Beta Limits

The W7-AS modular stellarator and the LHD exhibit hollow $n_e(r)$ profiles with a center/peak density ratio of 0.8 at low collisionality. Examples are shown in Fig. 11.

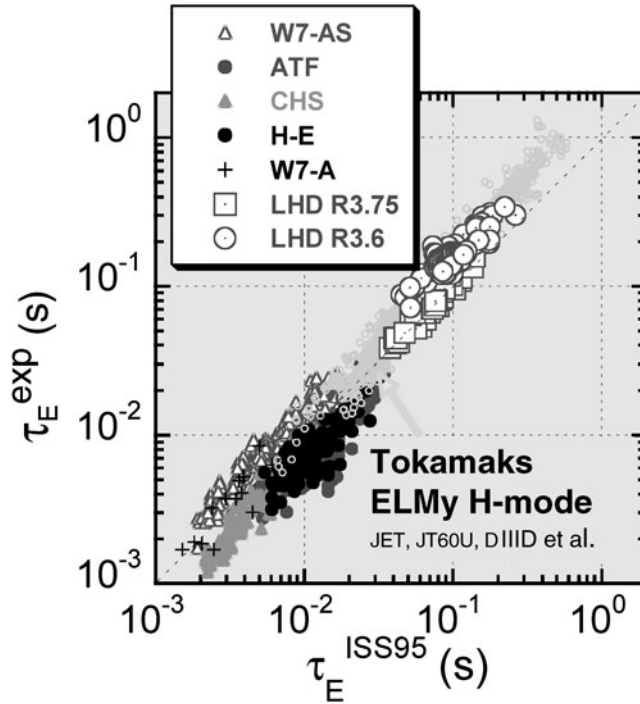


Fig. 9. Experimental energy confinement times versus the empirical fit.

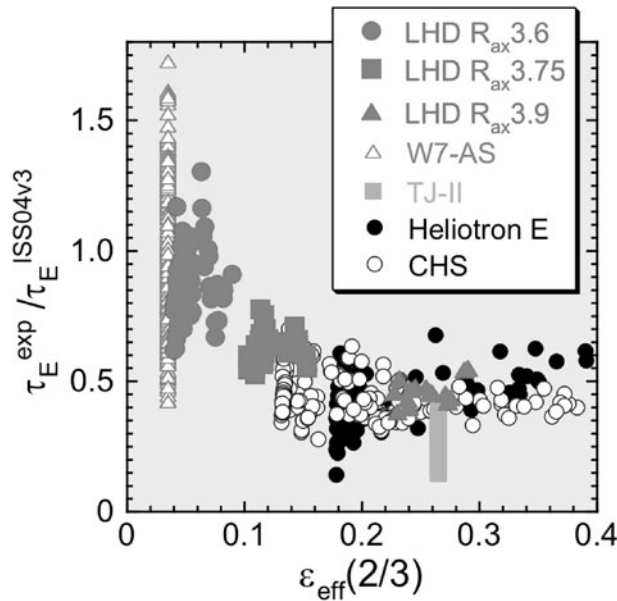


Fig. 10. Higher values of confinement improvement correlate with lower values of ϵ_{eff} .

The parameters for the LHD case are (a) 1-MW neutral beam injection (NBI), $T_i(0) = 1.3$ keV, and (b) 6.5-MW NBI, $T_i(0) = 1.9$ keV; for the W7-AS case the corresponding parameters are $T_e(0) = 1.5$ keV with electron cyclo-

tron heating. The expression used for the density in the ARIES-CS study is

$$n_e(r) = (n_{e0} - n_{edge}) \times \{[1 - (r/\langle a \rangle)^{expn}]^{\alpha n} \times [x_{midn} + (1 - x_{midn})(r/\langle a \rangle)^2]\} + n_{edge}, \quad (3)$$

where $expn = 12$, $\alpha n = 1$, $x_{midn} = 0.66$, and $n_{edge}/n_{e0} = 0.1$ for the reference case. This expression allows us to study a wide range of density profiles including hollow profiles and those used in other power plant studies. The value of $\tilde{n}_e$, the line-averaged density, is constrained to be $\leq 3\tilde{n}_{Sudo}$, where $\tilde{n}_{Sudo}$ (Ref. 15) is given by

$$\tilde{n}_{Sudo} = 0.25[P_{plasma}\langle B_{axis} \rangle / (\langle R_{axis} \rangle \langle a \rangle)^2]^{1/2}. \quad (4)$$

Values of (2 to 3.5) $\tilde{n}_{Sudo}$ have been obtained in LHD (Ref. 16). For the reference parameters $\langle R_{axis} \rangle = 7.75$ m, $\langle B_{axis} \rangle = 5.7$ T, $P_{heating} = 462$ MW (5% $f_{\alpha,lost}$ and $P_E = 1$ GW), the $\tilde{n} \leq 3\tilde{n}_{Sudo}$ requirement corresponds to $\tilde{n} \leq 8.12 \times 10^{20} \text{ m}^{-3}$, which is easily met for all cases of interest.

The temperature profile was chosen to give the same pressure profile as that used in the ARIES-CS magneto-hydrodynamic (MHD) studies,⁷ which in turn is the same as that adopted for the ARIES-RS tokamak power plant study.¹⁷ The ion and electron temperatures are assumed to be equal at the high densities appropriate for an ignited stellarator reactor. Figure 12 shows the radial profiles for density, temperature, and pressure for the reference case. The sensitivity to the shape of the density and temperature profiles is discussed in Sec. IX.D.

The scrape-off layer (SOL) equations are used in a modified two-point Borass model¹⁸ to determine the edge density and temperature needed to satisfy requirements on radiating 75% of the particle power flowing into the SOL and limiting the temperature at the divertor to ~ 10 eV to minimize sputtering of divertor material. The value for T at the divertor plate gives very low sputtering coefficients, which should give a divertor plate lifetime longer than the replacement time for the other replaceable components, including the blankets. Because of the nonlinearity of the temperature dependence of the SOL radiation, the solution is obtained computationally. Carbon is used in the calculation to obtain the proper radiation level but higher-Z materials could also be used. The calculations show that an operational window exists for the connection lengths found by following field lines to the divertor for an edge temperature ~ 200 eV and an edge density in the $3 \times 10^{19} \text{ m}^{-3}$ range. The reference edge values are $n_{e,edge}/n_{e,axis} = 0.1$ and $T_{edge}/T_{axis} = 0.002$ at $r/a = 1$ and 0.017 at $r/a = 0.98$. For the reference case $n_{edge} = 3.8 \times 10^{19} \text{ m}^{-3}$, $T(r/a = 0.98) = 197$ eV, and $T(r/a = 1) = 23$ eV, which satisfies the SOL radiation and divertor sputtering conditions.

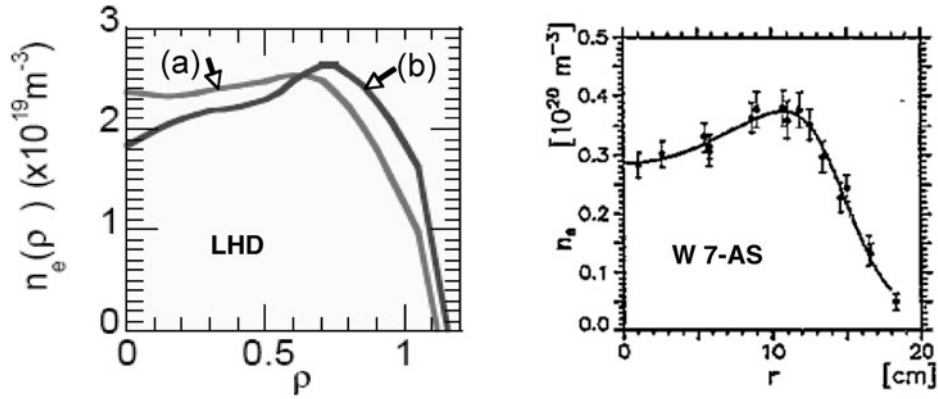
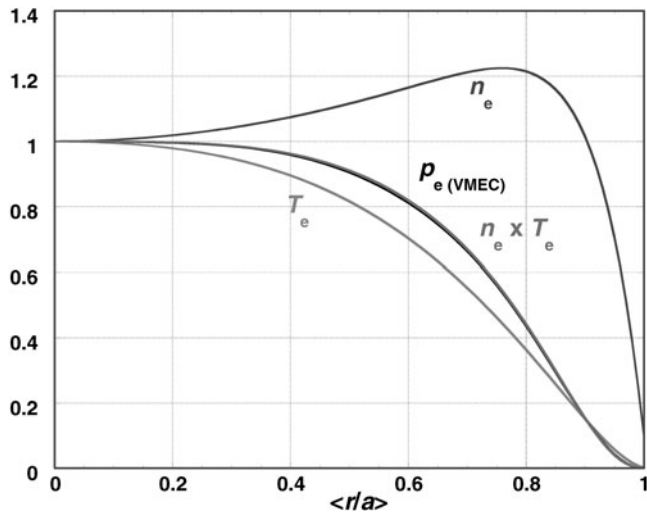
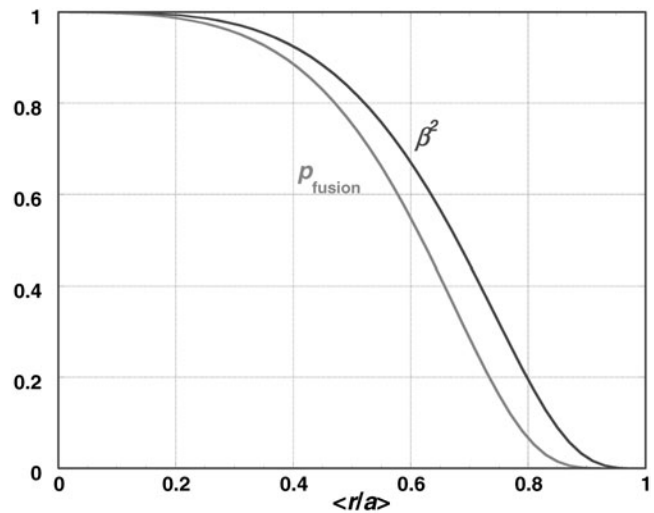

 Fig. 11. Hollow $n_e(r)$ profiles in LHD (left) and in W7-AS (right).


Fig. 12. Radial profiles used for the reference case.


 Fig. 13. Normalized profiles for the fusion power density and β^2 .

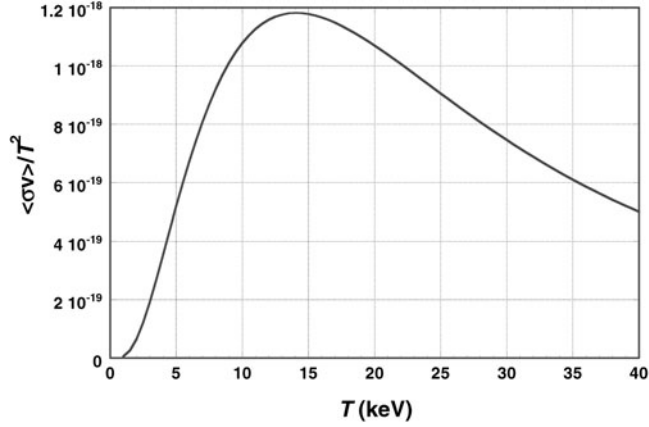
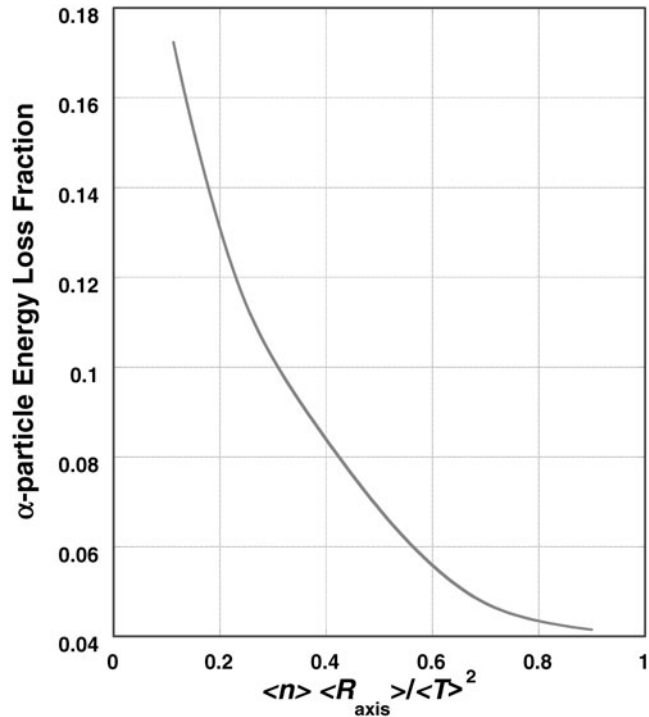
V.C. Other Conditions and Constraints

Figure 13 shows the normalized profiles for the reference case for the fusion power density $p_{\text{fusion}}(r/a) \propto n^2 \langle \sigma v(T) \rangle$ and $\beta^2(r/a) \propto (\langle n \rangle \langle T \rangle)^2$, which is broader than $p_{\text{fusion}}(r/a)$. These profiles would be the same if $\langle \sigma v(T) \rangle$ were proportional to T^2 , which is only approximately true for a range of temperatures, as shown in Fig. 14. In the range of interest, $\langle \sigma v \rangle$ varies approximately as $T^{2.5}$. Reference 7 analyzed the α -particle energy loss rate for different ARIES-CS plasma parameters. Figure 15 shows the dependence of this loss rate, which varies as $\langle R_{\text{axis}} \rangle^2 \langle B_{\text{axis}} \rangle / A_p^2$, for the reference case on $\langle n \rangle \langle R_{\text{axis}} \rangle / \langle T \rangle^2$, which is related to the plasma collisionality. The target value for the loss rate is $\leq 5\%$, which corresponds to $\langle n \rangle \langle R_{\text{axis}} \rangle / \langle T \rangle^2 \geq 0.72 \text{ m}^{-2} \cdot \text{keV}^{-2}$ or $\langle n \rangle \geq 4.01 \times 10^{20} \text{ m}^{-3}$ for the reference case. A $\langle \beta \rangle$ value of 6.4% is obtained for the reference case. There is

no good model for the beta limit in stellarators; values of $\langle \beta \rangle = 4.8\%$ have been obtained in LHD and it appears that the β limit is more determined by equilibrium quality than by instability in stellarators.¹⁹ The equilibrium $\langle \beta \rangle$ values for NCSX-type plasmas is significantly higher than 5% (Ref. 7).

V.D. Impurity Profiles, Core Radiation, and Radiation in the SOL

Proper treatment of impurities is important in assessing ARIES-CS plasma performance. The electron density $n_e(r) = n_{\text{DT}}(r) + 2n_{\text{He}}(r) + \sum Z n_Z(r)$, so high impurity levels reduce $p_{\text{fusion}} \propto n_{\text{DT}}^2 \langle \sigma v \rangle$ through reduced n_{DT} and reduced electron temperature T_e (and hence the D-T temperature) through higher radiative losses. Higher values of $\langle B_{\text{axis}} \rangle$, or confinement

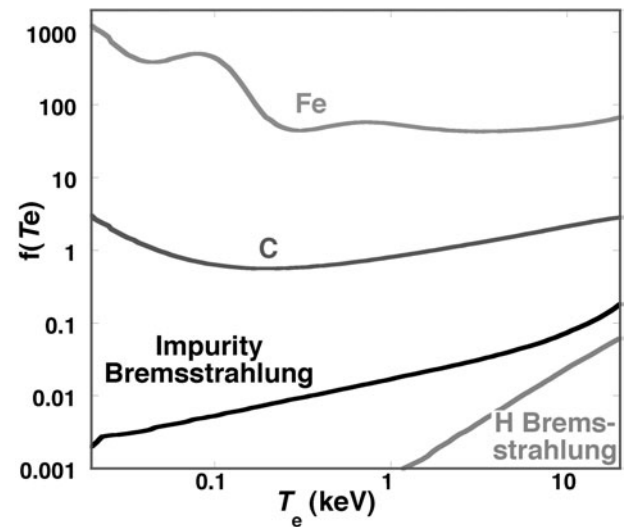

 Fig. 14. Fusion reactivity/ T^2 .

 Fig. 15. Alpha-particle energy loss fraction versus $\langle n \rangle \langle R_{axis} \rangle / \langle T \rangle^2$.

improvement factor H-ISS95, or $\langle R_{axis} \rangle$ are needed to compensate for higher impurity levels. The helium ash density is calculated from the fusion reaction rate, so $\langle n_{He} \rangle / \langle n_e \rangle = [P_{fusion} / (17.6 V_{pl})] (\tau_{He}^* / \tau_E) (\tau_E / \langle n_e \rangle)$; $\langle n_{He} \rangle / \langle n_e \rangle = 3.35\%$ for the reference case parameters with $\tau_{He}^* / \tau_E = 6$.

Iron is chosen for the plasma impurity species, and for the reference case we assume that the impurity density is a constant fraction of the electron density. The standard coronal model for line radiation and electron-

ion recombination is used for calculating the radiative power density $p_{radiation} \propto n_e n_Z f(T_e)$, where $f(T_e)$ is plotted in Fig. 16. Figure 17 shows the radiation power density profiles for the reference case. The total amount of power radiated ($P_{radiation}$) equals the bremsstrahlung (D-T + He + Fe) plus the Fe line radiation; the structure near the lower temperature edge is due to the Fe line radiation, as can be seen from Fig. 16. The fraction of iron in the plasma is chosen to give a 75% reference case target value for $f_{rad} = P_{radiation} / P_{plasma}$, where $P_{plasma} = P_\alpha (1 - f_{\alpha,loss})$ is the alpha-particle power to the plasma. The Fe radiation accounts for 62.4% of the total radiation, D-T accounts for 32.9%, and He accounts for 4.7%. Hollow $n_e(r)$, and hence hollow $n_Z(r)$, profiles are of interest because they produce broader radiation profiles, which reduce the ratio of the maximum-to-average radiative power density on the wall.¹¹ Since the average radiative wall power density is fixed for a given f_{rad} and $\langle R_{axis} \rangle$, reducing this ratio reduces the peak cooling needs on the wall.

Figure 18 illustrates the power flows within the plasma and to the first wall and divertor. Very close to 80% of the D-T fusion power goes to the wall as 14-MeV neutrons ($P_{neutron}$), and the remaining approximately 20% (P_α) as α particles that slow down in the plasma and transfer their energy to the plasma; a certain fraction $f_{\alpha,loss}$ (targeted at $\leq 5\%$ of P_α) is lost to the divertor. Energy multiplication in the blanket increases $P_{neutron}$ by 16%, and 90% of the power for helium pumping and balance of plant is returned as thermal power, which is converted with a 43% thermal conversion efficiency. Of the power that goes to the plasma, a fraction f_{rad} (targeted at $\geq 75\%$, similar to that in tokamak reactor studies²⁰) is radiated from the plasma and $(1 - f_{rad})$ leaves the plasma in


 Fig. 16. Temperature dependence of radiative power density, $p_{radiation} / n_e n_Z$.

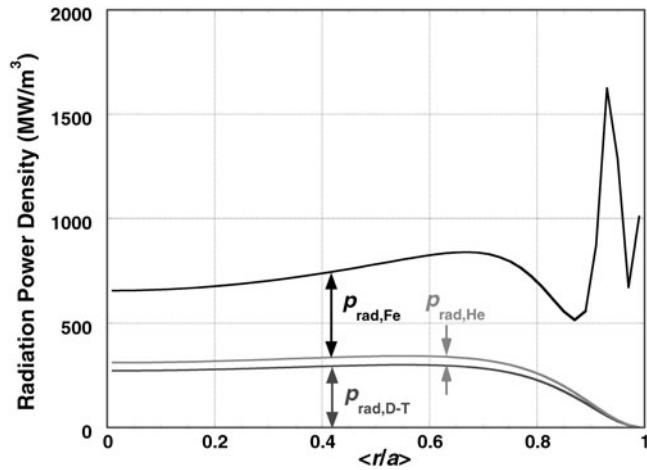


Fig. 17. Radiation power profiles.

thermal particles that cross the last closed flux surface and appear in the SOL. It is assumed that a certain fraction $f_{\text{SOL}} = 75\%$ of this power is radiated in the SOL, so $P_{\text{plasma}}(1 - f_{\text{rad}})(1 - f_{\text{SOL}})$ hits the divertor plates as thermal particles (in addition to the lost α particles). The model assumes that half of this SOL radiation is to the wall and half occurs in front of the divertor. Of the SOL radiation that occurs in front of the divertor, it is assumed that half goes to the wall and half to the divertor plates and baffles. These fractions are similar to those assumed for tokamaks. The net result is that approximately equal amounts of power impact the divertor as α particles and

thermal particles, so $P_{\text{div}} \approx 0.1P_{\alpha}$. The effect of changing the underlined percentages in Fig. 18 is discussed in Sec. IX.C.

The large fraction of α -particle power radiated to the wall means that less power exits the plasma in particles into the SOL, a fraction (0.25 in the reference case) of which impacts on the divertor. Only a small fraction of the core radiation impacts the divertor. While the divertor is assumed to cover 10.6% of the first-wall area, the fraction of the radiated power incident on the divertor is less because of the spatial distribution of the radiated power on the wall.¹¹ The divertor is in the area close to where the minimum radiated power density occurs; $P_{\text{rad}}(\text{divertor location})/P_{\text{rad}}(\text{average}) \approx 0.5$. This corresponds to an effective fractional wall coverage $f_{\text{div,eff}}$ of $\approx 5.3\%$. The total power radiated to the first wall is due to radiation only from the core and that in the SOL:

$$P_{\text{wall}} = P_{\alpha}(1 - f_{\alpha,\text{loss}}) \times \{f_{\text{rad}}(1 - f_{\text{div,eff}}) + (1 - f_{\text{rad}}) \times f_{\text{SOL}}[0.25 + 0.5(1 - f_{\text{div,eff}})]\} \quad (5)$$

For the reference values $f_{\alpha,\text{loss}} = 0.05$, $f_{\text{rad}} = 0.75$, $f_{\text{div,eff}} = 0.053$, and $f_{\text{SOL}} = 0.75$, $P_{\text{wall}}/P_{\alpha} = 0.80$. For a wall area (not including the divertor) = 651 m^2 , the average radiated power density on the wall plus divertor is 0.54 MW/m^2 . The peak-to-average power ratio for radiation on the wall (1.4) is such that it does not present the same problem as the power in particles incident on the divertor.

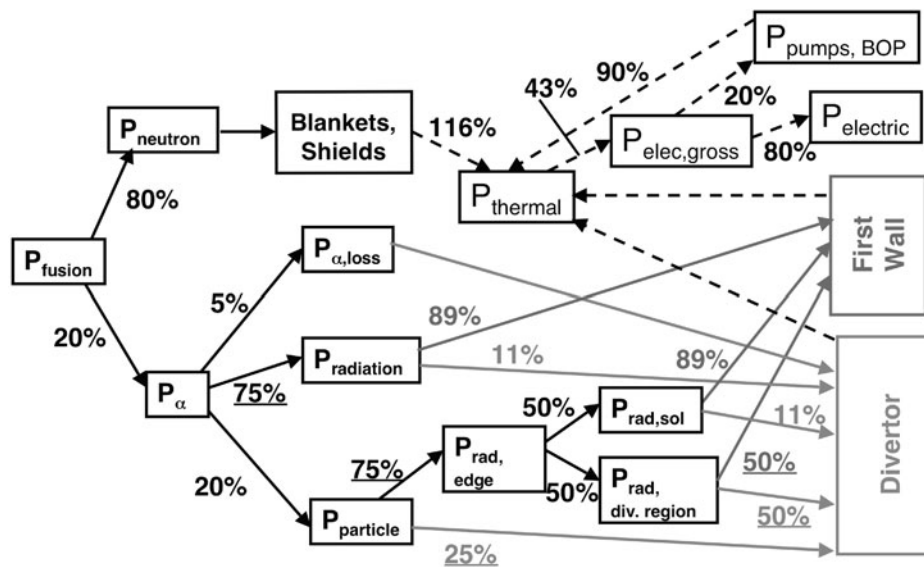


Fig. 18. Plasma and component power flows.

The total power to the divertor is

$$P_{divertor} = P_{\alpha} \{ f_{\alpha, loss} + (1 - f_{\alpha, loss}) \times [(1 - f_{rad})(1 - f_{SOL}) + f_{rad} f_{div, eff} + (1 - f_{rad}) f_{SOL} (0.5 f_{div, eff} + 0.25)] \} . \quad (6)$$

For the reference values $P_{divertor}/P_{\alpha} = 0.20$. The terms $f_{\alpha, loss} + (1 - f_{\alpha, loss})(1 - f_{rad})(1 - f_{SOL}) = 0.109$ in Eq. (6) represent the power in α particles and thermal particles, respectively, that impact the divertor. The power impacting the divertor in each group of particles is approximately the same (5 and 5.9%, respectively); 5% each was assumed in the divertor power calculations in Ref. 21. Minimizing the power to the divertor is important in limiting the peak power density on the divertor, which is already above the desired limiting value. The effect of varying the reference power fractions is discussed in Sec. IX.C.

VI. COSTING MODEL

The ARIES-CS cost model is similar to that used in earlier ARIES studies²² but updated for the ARIES-CS geometry and the relevant component compositions. All costs are for complex machined shapes in year 2004 dollars, corresponding to the start of the study. Costs are translated to year 1992 dollars only in Sec. XI, where comparisons are made with some ARIES power plant studies done in the 1990s.

VI.A. Reactor Core Geometry and Unit Cost Values

Figure 5 shows the structure and nominal radial dimensions of the components between the plasma edge and the modular coils. The composition, thickness, and percentage of coverage (discussed in Ref. 11) and the density and cost/kg for each component are given in Table II. The coil support structure (the thin cover over coils that faces the plasma, the intercoil structure, and the strongback radially behind the coils) is an advanced design²³ with a low fabrication cost, and its thicknesses (obtained from stress calculations¹²) are scaled by $((B_{axis})/R_{axis})^2$ to keep the stress in the coil structure the same as in the 3-D finite element analysis ANSYS calculations. The sensitivity of the cost of the coil structure and the other main components (first wall, blankets, shield, vacuum vessel, and modular coils) to individual cost multipliers due to fabrication risk is discussed in Sec. X.B. The values in Table II are used to relate the component volumes to masses and costs. The volumes of these components are calculated from their average thickness and

the area through their midpoint scaled from the area of the last closed flux surface by the ratio of the distance to the midpoint over the average plasma radius. This estimate is close to that obtained from the 3-D ANSYS calculation of the volumes and allows scaling with major radius and neutron wall power density. In addition, the length (and hence volume and cost) of the winding pack, the plasma-facing coil cover, and the strongback are multiplied by a factor 1.135 to agree with the 3-D finite element analysis ANSYS model of the coils.

VI.B. Cost Accounts

The cost accounts and most of the costing algorithms are the same as those used in earlier ARIES studies.²² Table III lists the cost accounts used in this study. The total direct cost for the reference case is 2620 million U.S. dollars, hereafter denoted by M\$ (other cost multipliers raise the total capital cost to 4898 M\$). Account 20 (land and land rights) is a small fixed cost, only 0.55% of the total direct cost. Account 23 (turbine plant equipment) is 12.6% of the total direct cost and includes turbine generators, the main steam system, condensing system, feed heating system, other turbine plant equipment, and instrumentation and control. Account 24 (electric plant equipment) is 5.7% of the total direct cost and includes switchgear, station service equipment, switchboards, protective equipment, electrical structures and wiring containers, and electrical equipment. Account 25 (miscellaneous plant equipment) is 2.9% of the total direct cost and includes transportation and lifting equipment, air and water service systems, communications equipment, and furnishings and fixtures. Account 26 (special materials) is 6.1% of the total direct cost and consists of the 70%-enriched LiPb coolant and breeder. Account 27 (heat rejection) is 2.3% of the total direct cost. The only cost accounts that depend on the specific compact stellarator configuration are accounts are 22.1.1 through 22.1.6 and 26, which account for 33.6% of the total direct cost. The plasma start-up power (cost account 22.1.4), assumed to be 20 MW of rf power, is 1.2% of the total direct cost. The other cost accounts depend only on the net electrical power produced and are independent of other device parameters.

VI.C. Coil Costing

The cost of the modular field coils is the sum of the cost of the material in the winding pack, which is 95% of the coil cost, and the cost of winding the conductor. The cost of the winding pack conductor is derived from the total length of the modular coils, their cross-sectional area (which gives the values for j and B_{max}), and the conductor cost per kA-m versus B_{max} from Fig. 8. The results are the same as the values for the reference case given in Ref. 12 but allow scaling with size and

TABLE II
Core Component Characteristics for the LiPb/FS/He Configuration

Component	Radial Depth (cm)	Area Fraction (%)	Composition (vol%)	Density (kg/m ³)	Unit Cost (2004 \$/kg)
First wall ^a	3.8	89.4	34% ferritic steel structure 66% helium coolant	7 800	103
Full blanket ^a	54.3	65.4	79% LiPb enriched 70% 7% SiC inserts 6% ferritic steel structure 8% helium coolant	8 897 3 200 7 800	17.1 101 103
Divertor system ^a	20	10.6	32.6% ferritic steel structure 4% W plate 63.4% helium coolant	7 800 19 300	103 105
Blanket behind divertor ^a	35	10.6	75% LiPb enriched 70% 9% SiC inserts 8% ferritic steel structure 8% helium coolant	8 897 3 200 7 800	17.1 101 103
Tapered blanket ^a	25 to 54.3	24	76% LiPb enriched 70% 8% SiC inserts 8% ferritic steel structure 8% helium coolant	8 897 3 200 7 800	17.1 101 103
Back walls ^a	5	100	80% ferritic steel structure 20% helium coolant	7 800	103
Ferritic steel full shield	32	76	15% ferritic steel structure 10% helium coolant	7 800	78
Ferritic steel tapered shield	0 to 14	24	75% borated steel filler	7 800	31
Tapered WC shield	26 to 34	24	15% ferritic steel structure 10% helium coolant 75% WC filler	7 800 15 500	78 30
Manifolds	35	80	52% ferritic steel structure 22.7% LiPb enriched 70% 24% helium coolant 1.3% SiC inserts	7 800 8 897 3 200	78 17.1 101
Vacuum vessel	28	100	28% ferritic steel structure 49% water 23% borated steel filler	7 800 7 800	56 31
Coil cover	2	28			10
Strongback outside coils	28		95% JK2LB steel structure 5% liquid helium coolant	7 800	28.9
Intercoil structure	16 to 28				28.9
Cryostat	5	100	100% Type 304 stainless steel	7 800	38.9

^aReplaced components.

field. The cost of winding the coils is taken from the same reference. The cost of the modular coil structure, with thicknesses estimated from the allowable stress in the structure,¹² is calculated from the values in Table II. The cost of the vertical field (VF) coil set including associated structure is obtained from an algorithm that takes into account the magnetic field on the VF coils,

the current in those coils (nominally zero in operation but set at 5 MA per coil for flexibility in plasma start-up), the current density in the coils (allowing for the volumes needed for copper, conductor sheath, and helium cooling at 4.2 K in addition to the NbTi strands), the resulting cost of the conductor and structure, and ancillary systems (vacuum, current leads, sensors).

TABLE III
Cost Accounts in Year 2004 M\$

20 Land and land rights	12.929
21 Structures and site facilities	336.133
21.1 Site improvements and facilities	22.840
21.2 Reactor building	136.642
21.3 Turbine building	43.757
21.4 Cooling system	10.307
21.5 Power supply building	12.371
21.6 Miscellaneous buildings	103.346
21.7 Ventilation stack	2.445
22 Reactor plant equipment	1538.817
22.1 Total core equipment	864.700
22.1.1.1 First wall	6.490
22.1.1.3 Blankets + back wall	52.856
22.1.1 Blankets, first, and back walls	59.347
22.1.2 Shield + back wall + manifold	228.627
22.1.3.1 Modular coils	115.960
22.1.3.2 VF coils	13.358
22.1.3.3 Divertor	5.318
22.1.3.4 Modular coil structure	93.535
22.1.3 Coils + structure	222.884
22.1.4 Plasma heating	66.427
22.1.5 Primary structure and support	73.126
22.1.6 Vacuum system and cryostat	137.135
22.1.7 Power supplies	70.624
22.1.8 Impurity control system	6.561
22.1.9 Direct energy conversion	0.000
22.1.10 Electron cyclotron heating start-up	0.000
22.2 Main heat transport system	474.771
22.2.1 Primary coolant loop	388.838
22.2.2 Intermediate coolant loop	0.000
22.2.3 Secondary coolant loop	85.933
22.3 Auxiliary cooling system	3.735
22.4 Radioactive waste treatment	6.655
22.5.1 Fuel injection	14.134
22.5.2 Fuel processing system	16.590
22.5.3 Fuel storage	7.067
22.5.4 Atmospheric tritium recovery	3.353
22.5.5 Water detritiation system	7.067
22.5.6 Blanket tritium recovery	7.067
22.5 Fuel handling and storage	55.279
22.6 Other plant equipment	60.723
22.7 Instrumentation and controls	44.558
23 Turbine plant equipment	314.558
24 Electric plant equipment	138.764
25 Miscellaneous plant equipment	70.958
26 Special materials	151.327
27 Heat rejection	56.086

VI.D. Calculation of the Levels of Safety Assurance Credits and the CoE

Following Miller et al.²² (and references therein), cost savings are assumed to arise from the substitution of

conventional (nonnuclear) components, as well as the elimination of certain active safety systems or other excess components, under the assumption of their inherent safety, as was done in previous ARIES studies. These cost credits represent both simplifications resulting from the elimination of active safety systems as well as reduction in costs associated with the quality assurance requirements mandated in the United States. The Levels of Safety Assurance (LSA) factors provide progressive discounts relative to nuclear-safety-grade costs for certain cost accounts. The reference LiPb/FS/He case assumes LSA = 2 where safety is assured by passive mechanisms as long as severe reconfiguration of large-scale geometry is not needed. The assumed LSA cost-credit multipliers²² are 0.95 for the blankets, shields, and coils, 0.90 for reactor and hot-cell buildings, 0.67 for other structures and improvements, 0.90 for heat transfer and transport, 0.94 for other reactor plant equipment, 0.84 for electrical plant equipment, and 0.90 for miscellaneous plant equipment. All other direct cost accounts are unchanged. The ratio of the added indirect costs to direct costs is 0.93. The operations and maintenance costs include a factor 0.85 and the costs for decontamination and decommissioning (waste disposal), $C_{D\&D}$, includes 0.5 mills/kW(electric)h (in 1992 dollars).

The CoE optimization figure of merit is given by

$$\text{CoE} = \frac{(C_{cap} + C_{O\&M} + C_{replace} + C_{fuel} + C_{D\&D})}{8760P_{e,net}f_{avail}N_{FPY}}, \quad (7)$$

where C_{cap} is total capital cost (the total direct cost times a fixed factor = 1.93 to account for fixed fractions for construction services and equipment, home office engineering and services, field office engineering and services, the owner's cost, process contingency, project contingency, and interest and escalation during construction, as discussed in Ref. 22). The total direct cost is the sum of the cost accounts listed in Table III. In Eq. (7) $C_{O\&M}$ is the cost for operations and maintenance, $C_{replace}$ is the cost for periodic replacement of the first wall, front blankets, back wall, and divertors, C_{fuel} is the cost of the deuterium fuel, $P_{e,net}$ is the net electric power, f_{avail} is the plant availability fraction (85% from Ref. 24), and $N_{FPY} = 40$ is the number of full-power years of operation. The total capital cost accounts for 82% of the CoE while operations and maintenance accounts for 14% and the cost of periodically replacing the first wall, front blankets, back wall, and divertors accounts for 3.5%. The fuel and decontamination and decommissioning costs are small contributors to the CoE.

VII. SYSTEMS CODE MODELS AND RESULTS FOR THE REFERENCE ARIES-CS POWER PLANT

The constraints used in obtaining the reference ARE case were an ignited plasma ($P_{input} = 0$), $P_{electric} = 1$ GW,

$f_{\alpha,loss} \leq 5\%$, $H\text{-ISS95} \leq 3$, $f_{rad} \geq 75\%$, $\tilde{n}/n_{Sudo} \leq 3$, $TBR > 1.1$, $B_{max} \leq 16$ T, $p_{n,wall,max} < 5.4$ MW/m², $j_{coil}/j_{SC}(B_{max}) \leq 1$, and the minimum distance between the last closed flux surface and the coil center larger than that required for the elements that are required in that space. The fixed parameters were $\tau_{He}^*/\tau_E = 6$, $f_{div,eff} = 0.053$, $f_{SOL} = 0.75$, and the form for the density and temperature profiles discussed in Sec. V.B.

VII.A. Plasma and Device Parameters and Power Components

The main device and plasma parameters for the reference case are listed in Table IV. A fusion power = 2436 MW is required (in combination with a blanket energy multiplication factor = 1.16 and a net thermal conversion efficiency = 43%) to produce the targeted 1-GW net electric power. Part (253 MW) of the gross electrical power is required for helium pump power in the blanket (170 MW), helium pump power in the divertor (27 MW), and plant plus cryogenic plant power (55 MW). The B_{max} value is less than the 16-T limit constraint; it is limited instead by $j_{coil} = j_{SC}(B_{max})$, Ampere's law, and the other constraints discussed in Sec. IV. The minimum value of $\langle\beta\rangle$ that satisfies all the constraints is 6.4%. The value for $\langle n \rangle$ (4.01×10^{20} m⁻³) is higher than that in tokamaks, but only $1.5n_{Sudo}$, and $\langle T \rangle$ is only 6.55 keV (central $T = 11.83$ keV), which reduces the helium ash dilution, as a result of the high $\langle n \rangle$ and large plasma volume. The high $\langle n \rangle / \langle T \rangle^2$ value required to reduce the α -particle loss to the desired (5%) level (Fig. 15) on the divertor plates means that the optimum value for $\langle T \rangle$ is below the peak of the $\langle \sigma v \rangle$ curve in Fig. 14. A higher $\langle T \rangle$ would result in an unacceptably high α -particle loss on the divertor plates; a compensating higher value for $\langle n \rangle$ to keep the α -particle loss at an acceptable level would not give the target P_{fusion} value and allow the other constraints to be satisfied. The required confinement multiplier, $H\text{-ISS95} = 2.04$, is modest compared to that expected in compact stellarators and is less than that already achieved in some stellarator experiments.

The 5.41 MW/m² value for $p_{n,wall,max}$ is higher than desired. There are two ways to address this: a larger value for $\langle R_{axis} \rangle$, which increases the CoE, as discussed in Sec. X.C, or to move the wall farther from the plasma in the region where $p_{n,wall,max}$ occurs. The latter approach is feasible because the high- $p_{n,wall,max}$ region is at the large- R side,¹¹ where there is room for an expanded wall (see Fig. 4) and the blanket/shield thickness is thickest. A detailed discussion of the distribution of neutron wall loading is given in Ref. 11. Increasing this distance locally by 25 cm reduces $p_{n,wall,max}$ by 20% to a more acceptable 4.5 MW/m². The resulting reduction in shield thickness and associated cost benefits, such as replacement costs, would help to offset the increased cost due to the local larger wall area.

TABLE IV
Main Device and Plasma Parameters

Power plant parameters	
Net electric power (MW)	1000
Gross electric power (MW)	1253
Thermal power (MW)	2916
Fusion power (MW)	2436
Tritium breeding ratio	1.115
Device parameters	
Major radius (m)	7.75
Field on axis (T)	5.70
Maximum field on coils (T)	15.08
$j_{coil}/j_{max}(B_{max})$	1.00
Coil dimensions (m × m)	0.194×0.743
Maximum neutron wall load (MW/m ²)	5.41
Plasma parameters	
Stored plasma energy (MJ)	551
Volume averaged beta (%)	6.40
rms volume-averaged beta (%)	7.42
Volume-averaged density (10^{20} m ⁻³)	4.01
Line-average density/ n_{Sudo}	1.50
Density-averaged temperature (keV)	6.55
Central plasma temperature (keV)	11.83
Central ion density (10^{20} m ⁻³)	3.57
Central electron density (10^{20} m ⁻³)	3.84
Fraction fuel to electrons	0.931
Fraction carbon impurity	0
Fraction iron impurity	0.038%
Fraction helium	3.35%
Z effective	1.07
Power flows	
Confinement time, τ (s)	1.19
ISS-95 confinement multiplier	2.04
α -particle loss rate	5%
α heating power to plasma (MW)	462
Radiated power (MW)	365
Radiated power fraction	75%
D-T bremsstrahlung (MW)	116
He bremsstrahlung (MW)	16.6
Iron radiation (MW)	262
Synchrotron radiation (MW)	1.68
Power to SOL (MW)	97.0
Power to divertor (MW)	98.1

VII.B. Component Summaries

Table V lists the masses for the main fusion core systems and their components (illustrated in Fig. 5): the first wall, divertor, blankets, back wall, vacuum vessel, modular coils and structure, VF coils and structure, primary structure, and cryostat. The largest-mass components are the coil structure (3465 tonnes; the total modular coil mass including the support structure and winding pack is 4092 tonnes), the shield and its back wall (3280

TABLE V
Component Mass Summary in Tonnes

Total modular coil mass (winding pack)	627.2
Modular coil structure mass	3 465
Strongback mass	1 549
Coil cover mass	89.2
Intershell mass	1827
Total VF coil mass	
Coil + structure mass	4 092
Blanket, divertor, first- and back-wall mass	662.5
First-wall mass	66.33
Divertor mass	54.10
Front full-blanket mass	210.6
Front blanket back-wall mass	232.4
Second blanket mass	29.91
Tapered blanket mass	69.21
Shield mass and back wall	3 280
Ferritic steel shield mass	1816
Front tapered ferritic steel shield mass	108.9
Tapered shield back-wall mass	71.01
Back tapered WC shield mass	999.9
Penetration shield mass	283.8
Mass of manifolds	1 305
Total vacuum vessel mass	1 440
Full-blanket vacuum vessel mass	1 131
Tapered vacuum vessel mass	309.3
Mass of primary structure	2 909
Cryostat mass	1 333
Mass of core without LiPb	13 688

tonnes), and the primary support structure (18.4% of the core volume, or 2909 tonnes). Smaller mass components are the cryostat (1333 tonnes), the manifolds (1305 tonnes), and the first wall + divertor + blanket + its back wall (663 tonnes). The mass of the LiPb coolant/breeder in the power core components (3532 tonnes) is not included in the total dry weight of the fusion power core (13 688 tonnes). There is an additional factor of 1.5 mass for the LiPb coolant/breeder in the remainder of the power plant piping. The mass of the components that need to be replaced routinely (the first wall, divertor, blankets, tapered ferritic steel shield, and back wall) is 842 tonnes. The other components do not need to be replaced during the 40 full-power years of power plant operation.

Table VI lists the direct costs for the main fusion core systems and their components in year 2004 M\$ units. The most expensive part is associated with the modular coil system (116 M\$ for the coils and 93.5 M\$ for its associated structure for a total of 210 M\$). The relatively low cost for the massive coil structure is due to the advanced technique assumed for its fabrication.²³ Other expensive systems are the shield and associated back

TABLE VI
Component Cost Summary in Year 2004 M\$

Total modular coil cost	116.0
Modular coil SC cost	110.3
Modular coil winding cost	5.70
Modular coil structure cost	93.54
Strongback cost	42.52
Coil cover cost	0.85
Intershell cost	50.17
Total VF coil cost	14.06
Coil + structure cost	209.50
Blanket, divertor, first- and back-wall cost	64.66
First-wall cost	6.49
Divertor cost	5.32
Front full-blanket cost	20.47
Front blanket back-wall cost	22.74
Second blanket cost	2.91
Tapered blanket cost	6.73
Shield cost and back wall	131.67
Ferritic steel shield cost	67.01
Front tapered ferritic steel shield cost	4.02
Tapered shield back-wall cost	6.95
Back tapered WC shield cost	32.67
Penetration shield cost	21.03
Cost of manifolds	96.96
Total vacuum vessel cost	60.55
Full-blanket vacuum vessel cost	47.55
Tapered vacuum vessel cost	13.00
Cost of primary structure	61.42
Cryostat cost	61.70
Cost of core without LiPb	638.8

wall (132 M\$), manifold (97 M\$), the first wall + divertor + blanket + its back wall (65 M\$), cryostat (62 M\$), and the primary support structure (61 M\$). The cost of the LiPb coolant/breeder in the power core components (59 M\$) is not included in the total direct cost of the fusion power core (639 M\$). There is an additional factor of 1.5 cost for the LiPb coolant/breeder in the remainder of the power plant piping. The cost of the components that need to be replaced routinely is 75 M\$ for each time. The number of replacements required during operation is $14 - 1 = 13$, which is obtained by dividing the full-power lifetime (40 MW·yr) and the maximum neutron wall loading (5.4 MW/m²) by the first-wall end-of-life fluence limit (15.7 MWyr/m² for the reference LiPb/FS/He blanket and shield design). The sensitivity to increased cost for the various components is discussed in Sec. X.B. These direct costs must be multiplied by 1.93 to obtain the total capital costs, which still assumes the high interest rate for borrowed capital that was assumed in the ARIES 1990s studies. A more recent interest rate would reduce the CoE.

VIII. OPERATING POINT AND STARTUP PATH

Figure 19 shows the operating space for the reference ARIES-CS (ARE) case with $\langle R_{axis} \rangle = 7.75$ m and $\langle B_{axis} \rangle = 5.7$ T. The solid contours indicate the heating power input required to sustain a given point in the $\langle n \rangle$, $\langle T \rangle$ plane; the plasma heating contours are in 2-MW increments from 2 to 50 MW and in 50-MW increments from 50 to 200 MW. The operating point is the intersection of the 2.436-GW P_{fusion} curve with the thermally stable portion of the ignition contour ($P_{in} = 0$). Operating points on the left (lower $\langle T \rangle$) branch of the ignition curve are thermally unstable. The operating point parameters are $\langle n \rangle = 4.01 \times 10^{20} \text{ m}^{-3}$, $\langle T \rangle = 6.55 \text{ keV}$, $\langle \beta \rangle = 6.4\%$, $H\text{-ISS95} = 2.04$, $\langle f_{DT} \rangle = 93\%$, $\langle f_{He} \rangle = 3.35\%$, and $Z_{eff} = 1.07$. The minimum power path to the operating point in a tokamak would be through the saddle point, but since the saddle point lies in the region where $\tilde{n} > 3\tilde{n}_{Sudo}$, a different path is chosen. A convenient minimum-power path follows the $\tilde{n} = 3\tilde{n}_{Sudo}$ curve, then over to the desired operating point. At very low values of $\langle n \rangle$ and $\langle T \rangle$, where $P_{fusion} \ll P_{in}$, $\tilde{n}_{Sudo} \propto P_{in}^{0.5}$ and $\tilde{n} \leq 3\tilde{n}_{Sudo} = 0.378 P_{in} (\text{MW})^{0.5}$ for the reference $\langle R_{axis} \rangle$ and $\langle B_{axis} \rangle$ values. For 20-MW heating, $3\tilde{n}_{Sudo} = 1.69 \times 10^{20} \text{ m}^{-3}$. At higher values of $\langle n \rangle$ and $\langle T \rangle$, where $P_{in} \ll P_{fusion} \propto \langle n \rangle^2$, $\tilde{n}/\tilde{n}_{Sudo}$ is a function of only $\langle T \rangle$: $\tilde{n}/\tilde{n}_{Sudo} \propto \langle \sigma v(T) \rangle^{-0.5}$, which decreases as $\langle T \rangle$ increases toward ignition, allowing $\langle n \rangle$ to increase. At the 1-GW P_E operating point $\tilde{n}/\tilde{n}_{Sudo} = 1.5$. The allowed values for $\langle n \rangle$ and $\langle T \rangle$ are on the low- $\langle n \rangle$, high- $\langle T \rangle$ side of this nearly vertical curve in the $\langle n \rangle$, $\langle T \rangle$ plane. This path to an ignited operating point would require 37 MW of heating power

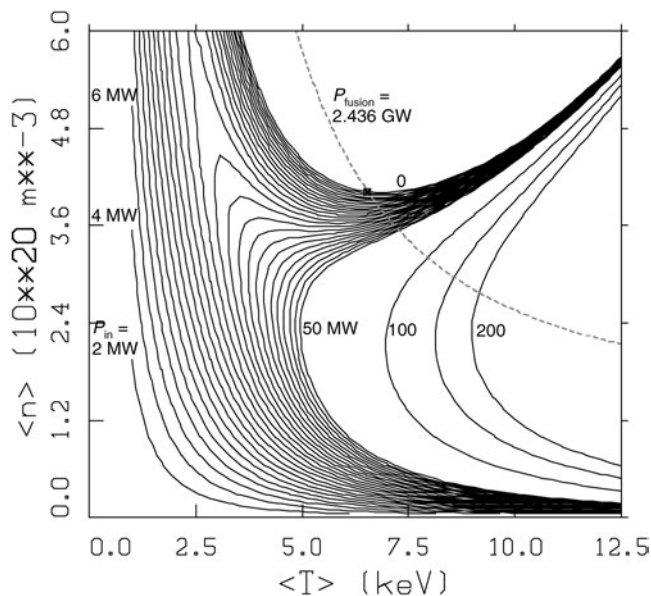


Fig. 19. Plasma operation contours for the base case.

with the minimum $H\text{-ISS95} = 2.04$ value for a stable operating point.

However, higher values for $H\text{-ISS95}$ are achievable during start-up, which lower the ignition contour, resulting in an operating point higher up the thermally stable (right) branch of the ignition contour and a lower value for the start-up power. Figure 20 shows the operating space for $H\text{-ISS95} = 2.4$ and 2.8, which allows startup powers of 13.5 and 5.4 MW, respectively. Since $H\text{-ISS95}$ values > 2.5 should be achievable, the start-up power can be reduced to < 10 MW, although 20 MW has been used in the costing. Figure 21 shows the variation of operating space parameters with the confinement multiplier. Points

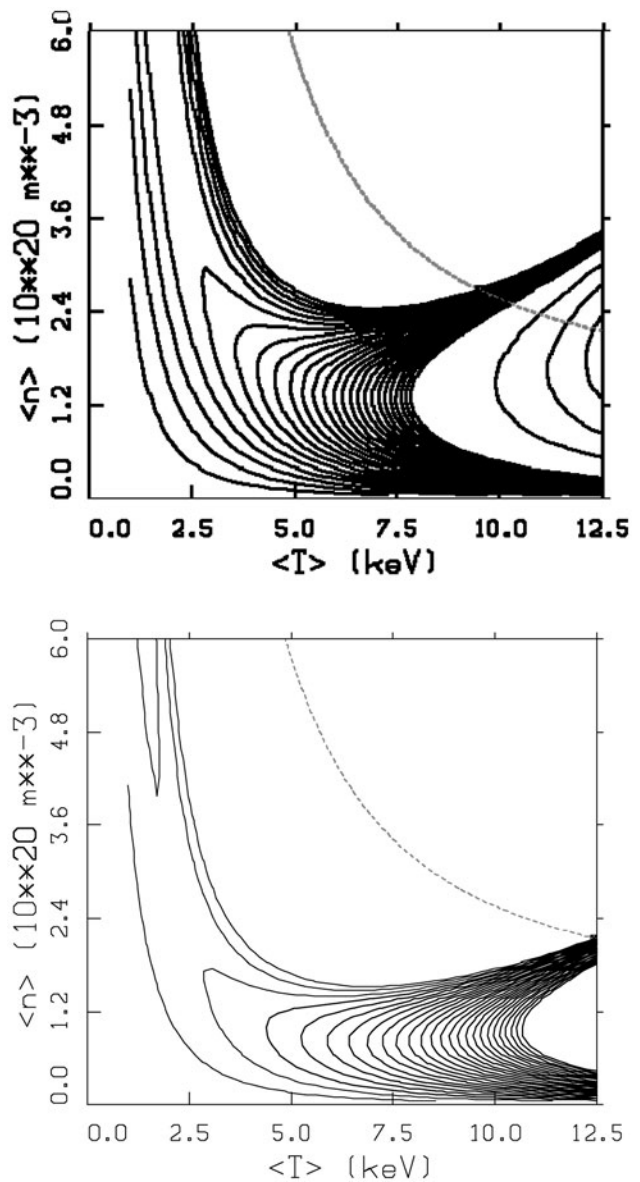


Fig. 20. ARIES-CS operating space for $H\text{-ISS95} = 2.4$ (top) and 2.8 (bottom).

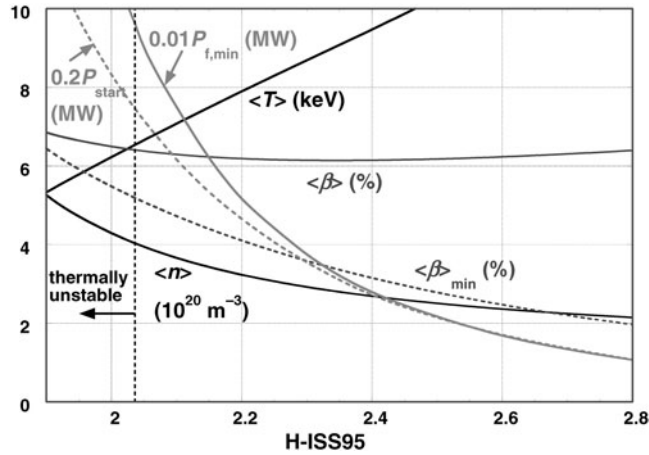


Fig. 21. Variation of operating space parameters with confinement multiplier.

to the left of the vertical dashed curve at $\text{H-ISS95} = 2.036$ are thermally unstable. The value of $\langle T \rangle$ varies linearly with H-ISS95 and $\langle n \rangle$ varies inversely with H-ISS95 such that there is only a small variation in $\langle \beta \rangle$ with H-ISS95 . The $P_{f,\min}$ and $\langle \beta \rangle_{\min}$ curves indicate the minimum fusion power along the ignition curve and the value of $\langle \beta \rangle$ at that point. For $\text{H-ISS95} = 2.4$, $P_{f,\min} = 277 \text{ MW}$ [$P_{\text{electric}}(\text{gross}) = 114 \text{ MW}$] and $\langle \beta \rangle_{\min} = 3.15\%$. For $\text{H-ISS95} = 2.8$, $P_{f,\min} = 107 \text{ MW}$ [$P_{\text{electric}}(\text{gross}) = 43.9 \text{ MW}$] and $\langle \beta \rangle_{\min} = 1.97\%$. Varying H-ISS95 allows operation at a lower fusion power level for licensing and commissioning before going to full-power operation, and it allows lower start-up power for normal operation.

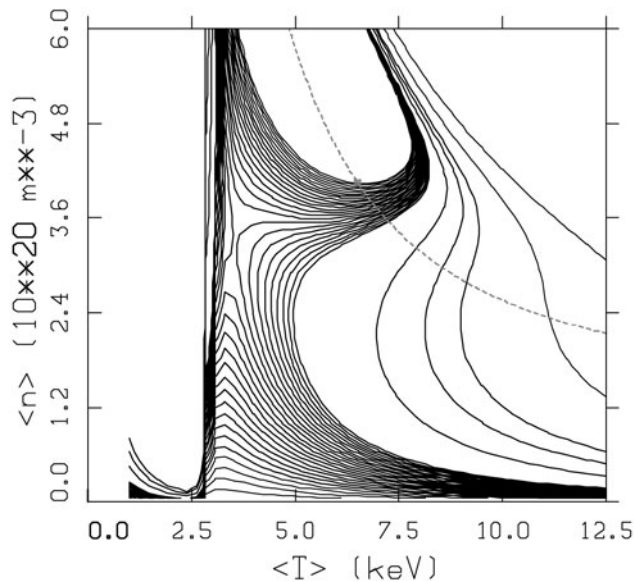


Fig. 22. Effect of confinement deterioration at high $\langle \beta \rangle$ and high $\langle n \rangle$.

The optimization and operating space calculations assume no additional deterioration of confinement time at high $\langle \beta \rangle$ or high $\langle n \rangle$. Figure 22 illustrates the change in the operating space when the confinement time is assumed to rapidly deteriorate with $\langle \beta \rangle$ beyond 12% and $\tilde{n}$ beyond $6n_{\text{Sudo}}$. The reduction in τ_E is assumed to be a factor of only 1.13 at 0.9 of each of these “limits” but rises to 1.43 at 0.95 of each limit, 2.72 at each limit, and 14.3 at 1.05 of each limit; the τ_E reduction factors for $\langle \beta \rangle$ and $\langle n \rangle$ are also multiplied. The effect is to collapse the left boundary at lower $\langle T \rangle$ and to compress the auxiliary heating curves at higher $\langle n \rangle$ and $\langle T \rangle$ such that the ignition curve becomes a closed loop in the $\langle n \rangle$, $\langle T \rangle$ plane. The differences due to this distortion of the operating space are relatively small as long as the ignition curve exists. For the same value of H-ISS95 as for the reference case, the differences (this case versus the reference case) are relatively minor: $\langle n \rangle$ (10^{20} m^{-3}) = 4.20 versus 4.01, $\langle T \rangle$ (keV) = 6.31 versus 6.55, and $\langle \beta \rangle$ (%) = 6.46 versus 6.40.

IX. BASE CASE VARIATIONS THAT AFFECT THE PLASMA PARAMETERS

The main device parameters for the reference case ($\langle R_{\text{axis}} \rangle$, $\langle B_{\text{axis}} \rangle$, B_{max} , and hence the CoE) and the resulting plasma parameters (primarily $\langle \beta \rangle$) were determined by the considerations discussed in Secs. III and IV. The reference plasma parameters were determined to satisfy a set of assumed and derived constraints and parameters: $\tilde{n}/n_{\text{Sudo}} \leq 3$, $f_{\alpha,\text{loss}} \leq 0.05$, $\text{H-ISS95} \leq 3$, $\tau_{\text{He}}^*/\tau_E = 6$, $f_{\text{rad}} \geq 0.75$, $f_{\text{SOL}} = 0.75$, a slightly hollow density profile, a temperature profile that gives the VMEC pressure profile used in the MHD stability studies,⁷ etc. It is instructive to examine the sensitivity of the overall optimization to these parameters.

IX.A. Density, Beta, and H-ISS95 Limits

Although there is a density limit $\tilde{n}/n_{\text{Sudo}} = 3$ imposed in the optimization, this limit was not reached in the parameter variations studied; typical values for $\tilde{n}/n_{\text{Sudo}}$ were ~ 1.5 . There is no hard constraint on the $\langle \beta \rangle$ value in the optimization. The $\langle \beta \rangle$ value is determined to be the minimum that satisfies all the constraints, which leads to a minimum value for $\tilde{n}/n_{\text{Sudo}}$. Higher limits would still yield the minimum values and lower limits would not satisfy all the constraints. The H-ISS95 value for the reference case is 2.04. The H-ISS95 values are typically ~ 2 , values that have been achieved in stellarators and in particular should be achievable in compact stellarators. The benefit of higher H-ISS95 is reducing the power needed for start-up to ignition, as discussed in Sec. VIII. The CoE is unaffected because it depends only on $\langle R_{\text{axis}} \rangle$, $\langle B_{\text{axis}} \rangle$, and B_{max} .

Forcing H-ISS95 to be 1.75, the lowest for which a solution exists, would require substantial increases in $\langle n \rangle$ (from $4.01 \times 10^{20} \text{ m}^{-3}$ to $7.58 \times 10^{20} \text{ m}^{-3}$), $\tilde{n}/n_{\text{Sudo}}$ (from 1.50 to 2.78), $\langle \beta \rangle$ (from 6.4 to 7.7%), and τ_E (from 1.19 to 1.39 s). The higher $\langle n \rangle$ value would cause decreases in $\langle T \rangle$ (from 6.55 to 4.15 keV), $f_{\alpha, \text{loss}}$ (from 5 to 2.04%), f_{He} (from 3.35 to 2.12%), and f_{Fe} (from 0.038% to 0.002%, the limiting reason why H-ISS95 cannot be lower with the assumed f_{rad} of 75%). However, this operating point is on the lower- $\langle T \rangle$ part of the ignition curve and is not thermally stable.

IX.B. Alpha-Particle Loss Fraction and Helium Accumulation

Figure 23 illustrates the sensitivity of the main plasma parameters to $f_{\alpha, \text{loss}}$, the fraction of the α -particle power that is lost; the base case has $f_{\alpha, \text{loss}} = 5\%$. From a power balance viewpoint, operating regimes with satisfactory parameters can be found for $f_{\alpha, \text{loss}}$ from 2 to 10%. The main benefit in reducing the α -particle power lost below 5% is to reduce the peak heat load on the divertor plates, which is already higher than desired.²¹ The other beneficial effects of a lower $f_{\alpha, \text{loss}}$ are lower helium dilution of the reacting fuel (due to $\tau_E/\langle n \rangle$ decreasing with lower $f_{\alpha, \text{loss}}$ where $\langle n \rangle$ is higher) and lower H-ISS95, neither of which is a concern. However, a lower $f_{\alpha, \text{loss}}$ for the NCSX configuration requires a higher value for $\langle n \rangle \langle R_{\text{axis}} \rangle / \langle T \rangle^2$ from Fig. 15, so $\langle n \rangle$ increases and $\langle T \rangle$ decreases, resulting in a higher $\langle \beta \rangle$. Nevertheless, these values are still acceptable and $f_{\alpha, \text{loss}} < 5\%$ is one route to reducing the peak heat load on the divertor.

The reference case assumes $\tau_{\text{He}}^*/\tau_E = 6$. Varying $\tau_{\text{He}}^*/\tau_E$ from 1.5 to 15 has a relatively small impact on the main plasma parameters except for the helium fraction in the plasma, which increases linearly with $\tau_{\text{He}}^*/\tau_E$ to 8.4%

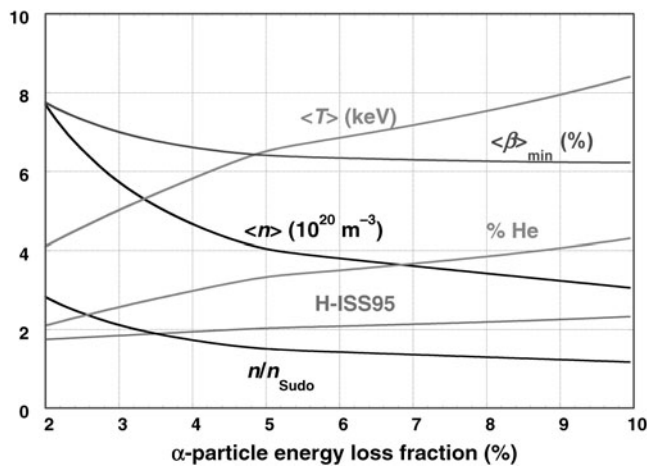


Fig. 23. Variation of plasma parameters with α -particle energy loss fraction.

at $\tau_{\text{He}}^*/\tau_E = 15$. The density and $\tilde{n}/n_{\text{Sudo}}$ increase by 18%, $\langle \beta \rangle$ increases by 13% over this range (which is not the same as the density increase because of the decrease in n_{ion}/n_e), and H-ISS95 falls by 4%. The value of $\langle T \rangle$ does not change over this range.

IX.C. Fractions of Power Radiated

The fraction of the power radiated from the plasma was varied from 23 to 95% with little effect on the plasma parameters: $\langle n \rangle$, $\tilde{n}/n_{\text{Sudo}}$, $\langle T \rangle$, and H-ISS95 varied $< 1\%$ from one extreme to the other. Only the amount of iron impurity needed for the radiated power target changed: $f_{\text{Fe}} = 10^{-6}$ at $f_{\text{rad}} = 23\%$ to 5.2×10^{-4} at $f_{\text{rad}} = 95\%$. The radiated power fraction cannot be $< 23\%$ because the D-T and helium bremsstrahlung set a limit for $f_{\text{rad}} = 23\%$, at which point no iron impurity is needed to obtain the radiated power fraction. Although increasing f_{rad} above 75% increases the radiative power load on the wall and divertor proportionately, this is not a significant concern because the peak power load on the wall is already below the limiting cooling value. The increase in f_{rad} from 75 to 95% increases the radiative power load on the wall by only 27% but reduces the power in thermal particles impacting the divertor plates by a factor of 2.5. The fraction of particle power radiated in the SOL (f_{SOL}) is assumed to be 75%. Values for f_{SOL} up to 80 to 90% are possible by impurity seeding in the SOL. Increasing f_{SOL} from 75 to 90% further reduces the power in thermal particles impacting the divertor plates by an additional factor of 2.5. While the divertor covers 10.6% of the wall area, the radiative power density in the divertor area is only half the average,¹¹ so the effective fractional divertor area for power consideration is $f_{\text{div, eff}} = 0.053$. Figure 24 shows the contours of constant P_{divertor} in the $f_{\text{rad}}, f_{\text{SOL}}$ plane with $f_{\text{div, eff}} = 0.053$ and $f_{\alpha, \text{loss}} = 0.05$.

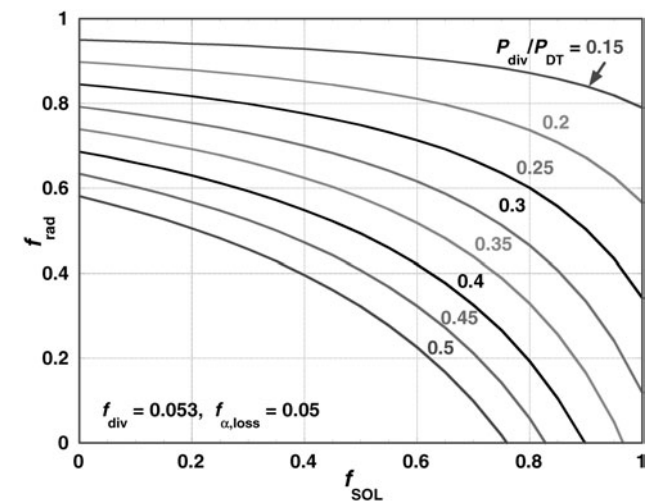


Fig. 24. Contours of constant P_{divertor} in the $f_{\text{rad}}, f_{\text{SOL}}$ plane.

The P_{div}/P_{DT} value for the reference values $f_{rad} = 0.75$ and $f_{SOL} = 0.75$ is 0.2. The minimum value for $P_{div}/P_{DT} = f_{\alpha,loss} + f_{div,eff} = 0.103$ corresponds to $f_{rad} = 100\%$. The reference value for f_{rad} is 75%, which corresponds to $P_{div}/P_{DT} = 0.20$. A combination of increasing f_{rad} and f_{SOL} above the reference values and reducing $f_{\alpha,loss}$ can make the divertor power-handling problem more manageable; the peak power density on the divertor exceeds the targeted 10 MW/m² by a significant amount for the reference case.¹²

IX.D. Density and Temperature Profiles

The form adopted for the pressure profile $p(r/a)$, shown in Fig. 12, is that used in the ARIES-RS tokamak power plant study. While this form for $p(r/a)$ leads to an adequate $\langle\beta\rangle$, it is not necessarily optimal for a compact stellarator in which the limiting $\langle\beta\rangle$ considerations seem to be flux surface deterioration rather than MHD stability. The sensitivity of the plasma parameters to the form for $p(r/a)$ was examined by varying the $n(r/a)$ profile using the general parametric form in Eq. (3) and a parametric form for the plasma temperature profile,

$$T(r/a) = (T_0 - T_{edge})\{[1 - (r/a)^x]^y\} + T_{edge} \quad (8)$$

Varying y from 0.5 to 3 for $x = 2$ (parabolic) and the reference (hollow) density profile leads to a pressure profile peaking factor $p(0)/\langle p \rangle$ varying from 1.44 to 3.83. Figure 25 shows the variation of the main plasma parameters with $p(0)/\langle p \rangle$ as the shape of the temperature profile, shown in Fig. 26, is changed; for the reference temperature profile derived from the VMEC $p(r/a)$, $p(0)/\langle p \rangle = 1.84$. All parameters increase with pressure profile peaking with $\langle\beta\rangle$ increasing the most. Pressure profiles

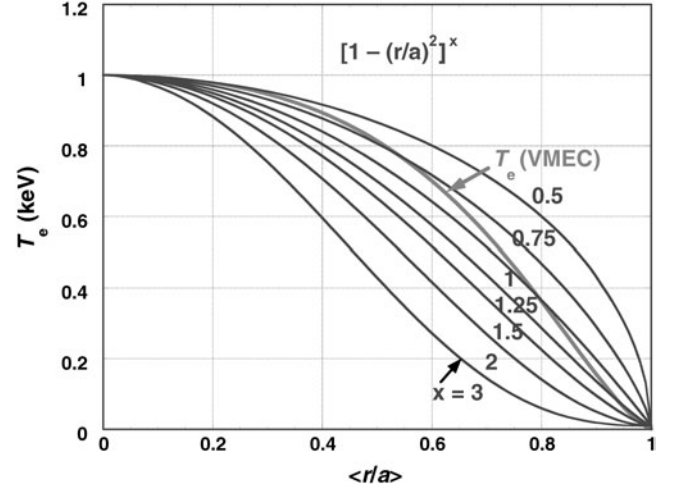


Fig. 26. Different temperature profiles.

less peaked than (parabolic)^{1,5} keep $\langle\beta\rangle$ below 6%. The pressure profile shape was also varied by varying the density profile, as shown in Fig. 27 with the $n(r/a)$ from Eq. (3), while keeping the reference VMEC-derived temperature profile. Figure 28 shows the resulting variation in plasma parameters. In this case all parameters decrease slowly with increasing $p(0)/\langle p \rangle$, except for $\tilde{n}/n_{Sudo}$. In both cases the plasma parameters are reasonably achievable, so the optimization results are relatively insensitive to the shape of the assumed pressure profile. The reference profiles require close to the highest $\langle n \rangle$, $\langle T \rangle$, $\langle\beta\rangle$, and H-ISS95, but these values are still moderate compared to those that can be expected in compact stellarators.

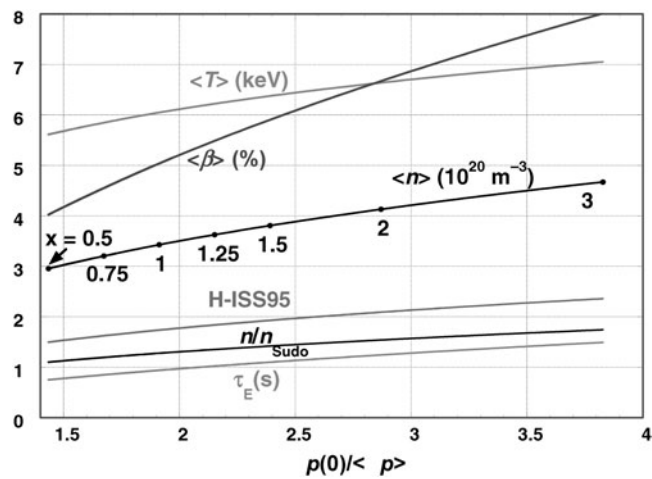


Fig. 25. Variation of plasma parameters with pressure profile peaking for the reference density profile.

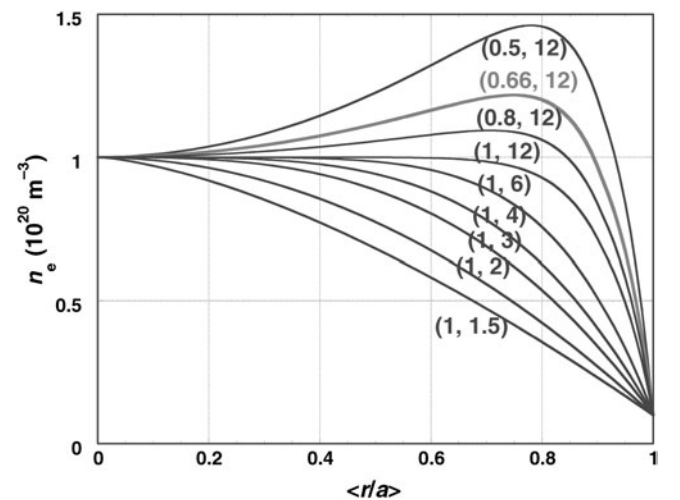


Fig. 27. Density profiles for different $(x_{midn}, expn)$ pairs.

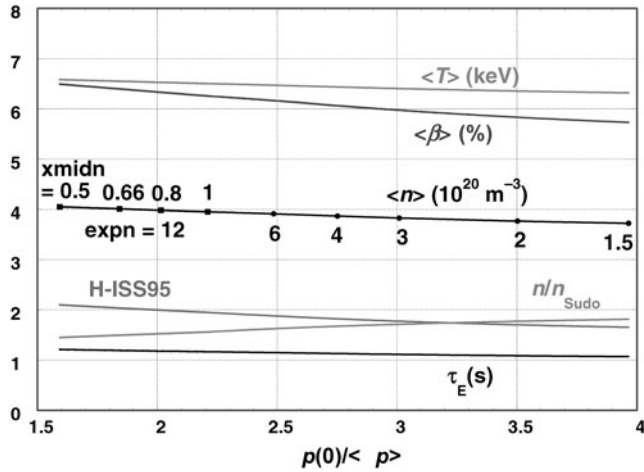


Fig. 28. Variation of plasma parameters with pressure profile peaking for the reference temperature profile.

X. VARIATIONS ABOUT THE REFERENCE CASE THAT AFFECT THE SIZE AND COST OF THE POWER PLANT

The reference ARIES-CS case is characterized by a set of assumed and derived choices for $p_{n,wall}$, B_{max} , component cost penalty, blanket and shield model, plasma and coil configuration, etc. It is instructive to examine the sensitivity of the overall optimization to these choices since these have the potential to affect the CoE, whereas the plasma parameter variations discussed in the previous section do not.

X.A. Maximum Magnetic Field on the Coils

Figure 29 shows the variation of various power plant parameters with B_{max} on the coils for a fixed $\langle R_{axis} \rangle =$

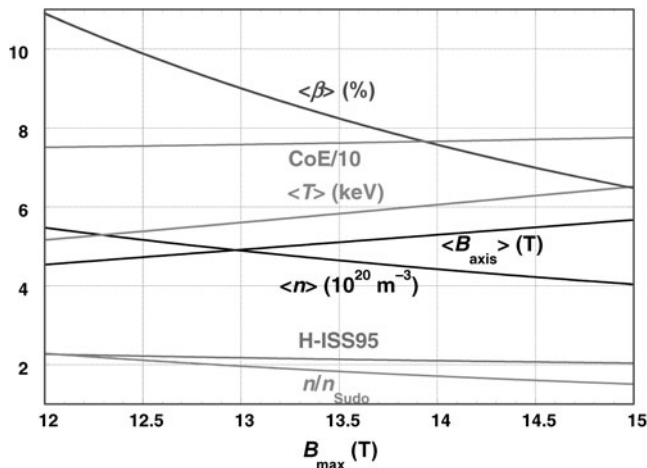


Fig. 29. Variation of reactor parameters with B_{max} .

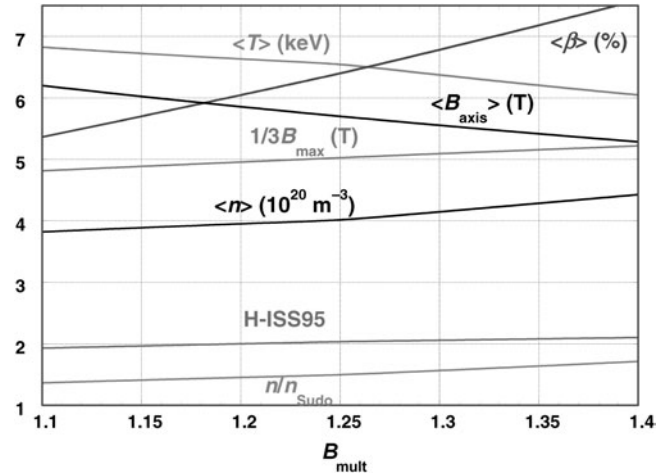


Fig. 30. Variation of reactor parameters with B_{mult} .

7.75 m. The $\langle n \rangle$ and $\tilde{n}/n_{Sudo}$ values decrease slowly with increasing B_{max} and $\langle T \rangle$ increases slowly with B_{max} such that the product $\langle n \rangle \times \langle T \rangle$ remains constant. The value of $\langle B_{axis} \rangle$ increases linearly with B_{max} and the required H-ISS95 decreases because of the $\langle B_{axis} \rangle$ dependence in τ_E^{ISS95} . The value of $\langle \beta \rangle$ increases with decreasing B_{max} as $1/\langle B_{axis} \rangle^2$. Although the CoE falls slightly with decreasing B_{max} , advantage cannot be taken of this reduction, such as use of NbTi for the modular coils, because of the much higher values of $\langle \beta \rangle$ required. The value of B_{max} calculated from the curves in Figs. 3 and 7 is multiplied by a constant B_{mult} to obtain a better fit to the 3-D ANSYS calculation of the maximum magnetic field at the surface of the winding pack. Figure 30 shows the effect of varying B_{mult} from 1.1 to 1.4 (the reference value is 1.25). The $\langle n \rangle$ and $\tilde{n}/n_{Sudo}$ values increase slowly with increasing B_{mult} and $\langle T \rangle$ decreases slowly with B_{mult} such that the product $\langle n \rangle \times \langle T \rangle$ again remains constant. The value of $\langle B_{axis} \rangle$ decreases linearly with B_{mult} and B_{max} increases.

X.B. Component Cost Penalties

The ARIES-CS reference case did not include a specific cost penalty for component complexity, other than that for complex machined shapes, and relied on advanced fabrication techniques for the massive coil support structure.²³ Using more conventional fabrication techniques increases the direct cost of the coil support structure from \$29/kg to \$56/kg in 2004 dollars. The net result would be an increase in the direct cost of the coil support structure from 93.5 M\$ to 180.5 M\$ and in the CoE from 78 to 80 mills/kW(electric)·h, again in 2004 dollar units. The optimization code allows an additional cost penalty factor for each major system (blankets, shielding, manifolds, vacuum vessel, coils, and coil structure). Applying a 25% cost penalty

to each major component separately increases the CoE by only 0.97 mill/kW(electric)·h for the blankets, 1.4 mills/kW(electric)·h for the shields, 1.3 mills/kW(electric)·h for the coils and support structure, and 0.01 mill/kW(electric)·h for the vacuum vessel. Combining them would increase the CoE by only 3.7 mills/kW(electric)·h. Quadrupling the complexity factor to 100% would increase the CoE by 14.7 mills/kW(electric)·h to 92.3 mills/kW(electric)·h, again in 2004 dollar units, or 72.9 mills/kW(electric)·h in the 1992 dollar units used in the earlier ARIES studies. By comparison, the CoE for the ARIES-RS tokamak power plant was 75.8 mills/kW(electric)·h in these same-year units.

It is difficult to judge what complexity factor to apply to the cost of winding the modular coils because the ARIES-CS case corresponds to a tenth-of-a-kind production reactor, when there would have been many decades of experience in building these types of coils, so the coil accuracy and fabrication techniques would not be the same as for today's first-of-a-kind modular stellarator coils. Since most of the coil cost is in the superconductor and the support structure, even applying a very large complexity penalty (e.g., a factor of 4) to the coil winding would only marginally increase the CoE and would not qualitatively change the favorable economic competitiveness of compact stellarators as a fusion power plant.

X.C. Scaling with $\langle R_{axis} \rangle$

The reference case was optimized for the lowest CoE, and hence $\langle R_{axis} \rangle$, and for the lowest $\langle \beta \rangle$. A larger value for $\langle R_{axis} \rangle$ could ease access constraints, reduce $p_{n,wall,max}$ and replacement costs, and improve the engineering design. Figure 31 shows the results of varying $\langle R_{axis} \rangle$ from 7.75 to 9.25 m. The value of B_{max} hits the

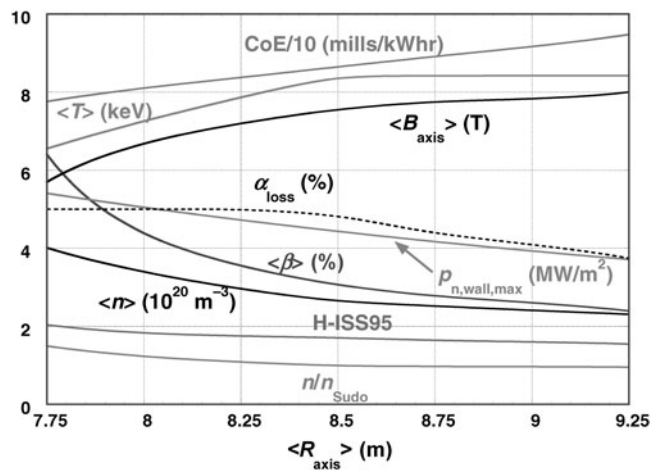


Fig. 31. Parameter variation with major radius for the reference plasma and coil configuration and the LiPb/FS/He blanket and shield concept.

16-T limit except for the $\langle R_{axis} \rangle = 7.75$ m case, where B_{max} is 15.1 T. All parameters decrease with increasing $\langle R_{axis} \rangle$ except for $\langle B_{axis} \rangle$, $\langle T \rangle$, and of course the CoE. The penalty associated with the increasing $\langle R_{axis} \rangle$ is 11 mills/kW(electric)·h·m⁻¹ for the reference LiPb/FS/He tapered blanket and shield concept, so it is of interest to examine other possibilities.

X.D. Full Blanket/Shield Case

The reference case assumes the full plus tapered blanket and shield geometry shown in Fig. 5. If a tapered blanket and shield region were not possible, then the blanket and shield would be similar to the full blanket and shield region at the left of Fig. 5 except that the blanket thickness is reduced by 4 cm because the reduced local TBR in the tapered region does not occur and the ferritic shield thickness is increased by 3 cm. The distance needed for the nominal 5-cm SOL, blanket, shield, manifolds, assembly gaps, vacuum vessel wall, plasma-facing coil structure, and 9.7-cm half-radial depth of the coil pack is 1.78 m; since $A_{\Delta} = 5.94$, this would give $\langle R_{axis} \rangle_{min} = 10.57$ m versus $\langle R_{axis} \rangle_{min} = 7.75$ m for the reference (tapered blanket and shield) case. However, the actual shield thickness is reduced by an additional 4 cm because the maximum neutron power density at the wall is reduced from 5.41 MW/m² to 3.08 MW/m² and the radial depth of the coil winding pack is reduced by 6.6 cm because the larger $\langle R_{axis} \rangle$ allows a larger toroidal extension of the winding pack. The net result is an increase in $\langle R_{axis} \rangle$ from 7.75 to 10.13 m and an increase in the CoE by 9.2 mills/kW(electric)·h (in 2004 dollar units). A further constraint was needed on $\langle B_{axis} \rangle$ to avoid too high a value for $\langle \beta \rangle$; $\langle B_{axis} \rangle$ was chosen to hold $\langle \beta \rangle$ close to that for the reference case. Less fusion power is required because of the increased thermal efficiency at the lower neutron wall power density, which reduces the pumping power required. The decrease in B_{max} from 15.1 to 12.4 T does not compensate for the higher component costs due to the larger wall area. The periodic cost of replacing the first wall, divertor, blankets, and back wall is more for the full-blanket case but fewer replacements are needed, resulting in a lower total cost for the replaced components. The other parameters for this case are compared in Table VII with those for the reference case, along with other cases to be discussed later in this section. The cost advantage of a tapered blanket and shield region is clear.

X.E. Advanced LiPb/SiC Blanket

The reference blanket and shield concept shown in Fig. 5 uses ferritic steel (FS) for structure, LiPb as breeder and coolant in the blanket, helium cooling in the blanket and shield, and water cooling on the vacuum vessel. An alternative is an advanced blanket and shield concept¹¹ (shown in Fig. 32) similar to that used for ARIES-AT

TABLE VII

Comparison of Different Blanket and Shield Configurations for a 1-GW(electric) ARE Power Plant

	Reference	ARE-1	ARE-2	ARE-3
Power plant parameters				
Blanket/shield composition	LiPb/FS/He LSA 2	LiPb/FS/He LSA 2	LiPb/SiC LSA 1	LiPb/SiC LSA 1
Blanket/shield type	Tapered	Full thickness	Tapered	Full thickness
Fusion power (MW)	2436	2354	1633	1625
Gross electric power (MW)	1253	1207	1055	1055
$p_{n,max}$ (MW/m ²)	5.41 ^a	3.08	3.63	3.47
Thermal efficiency (%)	43.0	43.4	59.8	60.1
CoE (mills/kW(electric)·h)	77.6	86.8	60.2	60.5
Device parameters				
$\langle R_{axis} \rangle$ (m)	7.75	10.13	7.75	7.90
Wall area (m ²)	728	1236	728	756
$\langle B_{axis} \rangle$ (T)	5.70	4.64	5.64	5.17
B_{max} (T)	15.1	12.4	15.0	14.4
Plasma parameters				
$\langle n \rangle$ (10 ²⁰ m ⁻³)	4.01	2.69	3.59	3.83
$\tilde{n}/n_{sudo}$	1.50	1.69	1.64	1.89
$\langle T \rangle$ (keV)	6.55	6.44	6.10	5.65
Plasma stored energy (MJ)	551	812	459	482
Plasma volume (m ³)	444	991	444	470
$\langle \beta \rangle$ (%)	6.40	6.37	5.45	6.44
H-ISS95	2.04	2.06	2.14	2.21
$f_{\alpha,loss}$ (%)	5	5	5	5
τ_E (s)	1.19	1.82	1.48	1.57
Component costs (M\$)				
Blankets and divertor	59.3	104	108	120
Shield, back wall, manifold	229	356	174	185
Coils	116	95.0	107	91.6
Coil structure	93.5	180	86.7	78.3
Total core cost	865	1193	789	791
Heat transport	475	462	175	174
Reactor plant	1539	1857	1112	1113
LiPb breeder/coolant	151	257	128	140
Total direct cost	2620	3026	1995	2008
Replaced components	966	776	476	471

^aConstraint limit.

(Ref. 25) with SiC/SiC composite for structure, LiPb as breeder in the blanket, LiPb cooling in the blanket and shield, and water cooling on the vacuum vessel. The advantages of the LiPb/SiC concept over the reference LiPb/FS/He concept are as follows:

1. higher thermal conversion efficiency (60% versus 43% for the reference case)
2. elimination of the helium coolant and manifolds and the associated pumping costs (in account 22.2) and a 40-cm-thinner radial build in the full-blanket region
3. elimination of the helium access pipes and the associated neutron streaming problem

4. a discrete LiPb manifold that does not have a shielding function.

The LiPb/SiC concept allows reduction in costs for passive safety associated with LSA = 1 (as for coal plants) for certain cost subaccounts.²² For LSA = 1, safety is assured by passive mechanisms of release limitation for any accident sequence, and radioactive inventories and material properties preclude a fatal release regardless of the reactor's condition.²² The assumed LSA cost-credit factors are the same as for the reference (LSA = 2) case except for the following cost accounts: 0.90 (versus 0.95) for the blankets, shields, and coils, 0.60 (versus 0.90) for reactor and hot-cell buildings, 0.60 (versus 0.67) for other structures and improvements, 0.85 (versus 0.94)

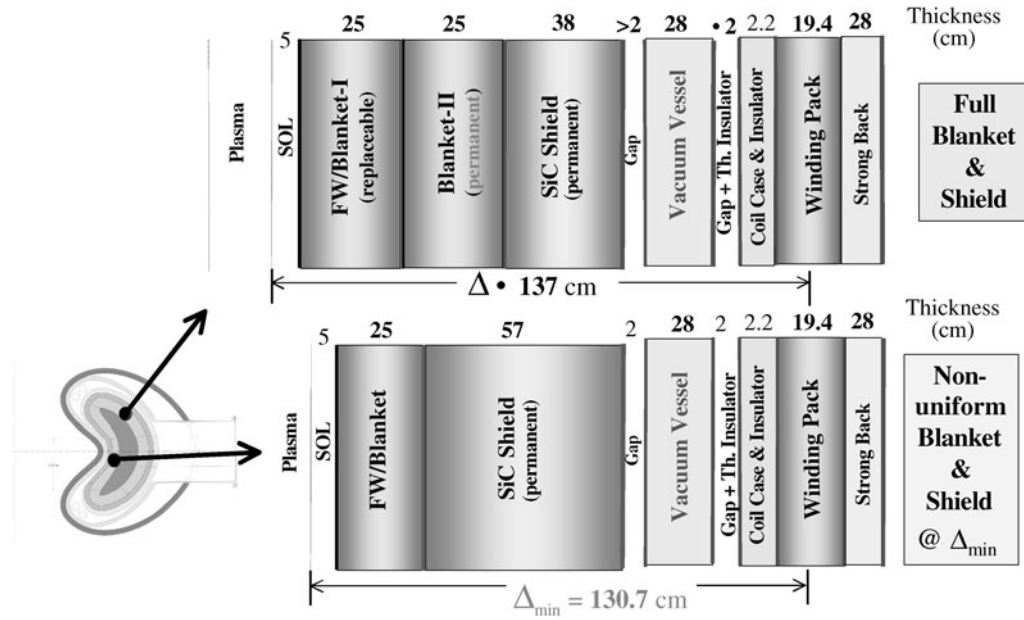


Fig. 32. Structure and nominal thicknesses for the LiPb/SiC blanket and shield concept. The first-wall end-of-life fluence limit is $18 \text{ MW} \cdot \text{yr}/\text{m}^2$, the overall TBR is 1.1, and the overall energy multiplication is 1.1.

for other reactor plant equipment, 0.75 (versus 0.84) for electrical plant equipment, and 0.85 (versus 0.90) for miscellaneous plant equipment. All other direct cost accounts are unchanged. The ratio of indirect costs to direct costs is 0.87 (versus 0.93). The operations and maintenance costs include a factor 0.70 (versus 0.85) and the decontamination and decommissioning allowance, $C_{D\&D}$, includes 0.25 mills/kW(electric)·h (in 1992 dollars) versus 0.50 mills/kW(electric)·h for the reference (LSA = 2) case.

The SiC/SiC structure is also lighter, resulting in a lower number of blanket modules and a shorter replacement time, which could improve the plant availability, but is more expensive (\$510/kg for SiC versus \$103/kg for ferritic steel in year 2004 dollars). The composition, thickness, and percentage coverage (discussed in Ref. 11) and the density and cost/kg (for complex machined shapes in year 2004 dollars) for each component are given in Table VIII.

The main device parameters are compared with those for the reference LiPb/FS/He blanket and shield in Table VII. The reduction in the peak neutron flux at the wall (from 5.41 to 3.63 MW/m^2 , due to the lower required P_{fusion} for the same net P_{electric}) does not allow a smaller $\langle R_{\text{axis}} \rangle$ because the minimum space required between the plasma edge and the center of the coil in the tapered region is the same as for the LiPb/FS/He blanket/shield case. However, the thickness of the blanket and shield is reduced a nominal 42 cm in the full-blanket region compared to that for the reference LiPb/FS/He blanket and shield in the corresponding region.

The fusion power required is much less because of the much higher thermal efficiency and elimination of the pumping power, as reflected in the reduced cost for heat transport. Only the 50 MW needed for the balance of plant power and 5 MW for cryogenic cooling enters into the gross electric power. The cost reductions due to LSA = 1 as well as the reduced shield thickness and lower thermal power handling lead to a much reduced CoE (from 78 to 60 mills/kW(electric)·h). Although the periodic cost of replacing the first wall, divertor, blankets, and back wall is more for the LiPb/SiC case, fewer replacements are needed, resulting in a lower total cost for the replaced components. There is a very large potential gain if the LiPb/SiC blanket and shield can be assumed, as it was in the ARIES-AT tokamak power plant study.²⁵

The large reduction in the CoE with the LiPb/SiC blanket and shield would allow a larger value for $\langle R_{\text{axis}} \rangle$, which could ease access constraints, reduce $p_{n,\text{wall,max}}$ and replacement costs, and improve the engineering design. Figure 33 shows the results of varying $\langle R_{\text{axis}} \rangle$ from 7.58 to 9 m. The value of B_{max} hits the 16-T limit at $\langle R_{\text{axis}} \rangle = 8$ m. All parameters decrease with increasing $\langle R_{\text{axis}} \rangle$ except for $\langle B_{\text{axis}} \rangle$, $\langle T \rangle$, and of course the CoE. The lowest values of $\langle R_{\text{axis}} \rangle$ have too high a $\langle \beta \rangle$ value, so a reasonable cutoff value for $\langle R_{\text{axis}} \rangle$ is 7.62 m, where $\langle \beta \rangle = 8.1\%$. The $\langle R_{\text{axis}} \rangle$ value corresponding to the reference $\langle \beta \rangle = 6.4\%$ is 7.7 m. The penalty associated with the increasing $\langle R_{\text{axis}} \rangle$ is 8.9 versus 11 mills/kW(electric)·h·m⁻¹ for the reference LiPb/FS/He tapered blanket and shield concept. Even the largest- $\langle R_{\text{axis}} \rangle$ (9-m) case

TABLE VIII
Core Components Characteristics for the LiPb/SiC Blanket/Shield Concept

Component	Radial Depth (cm)	Area Fraction (%)	Composition (vol%)	Density (kg/m ³)	Unit Cost (2004 \$/kg)
First wall and full blanket ^a	25	65.4	21% SiC/SiC structure 79% LiPb enriched 70%	3 200 8 897	510 17.1
Second blanket	25	65.4	21% SiC/SiC structure 79% LiPb enriched 70%	3 200 8 897	510 17.1
Divertor system ^a	20	10.6	33% SiC/SiC structure 4% W plate 63% LiPb enriched 70%	3 200 19 300 8 897	510 105 17.1
Blanket behind divertor	25	10.6	21% SiC/SiC structure 79% LiPb enriched 70%	3 200 8 897	510 17.1
Tapered blanket ^a	25 ^a to 50	24	21% SiC/SiC structure 79% LiPb enriched 70%	3 200 8 897	510 17.1
SiC full shield	38	76	15% SiC/SiC structure	3 200	510
SiC tapered shield	38 to 57	24	10% LiPb enriched 70% 75% borated steel filler	8 897 7 800	17.1 31
Vacuum vessel	28	100	28% ferritic steel structure 49% water 23% borated steel filler	7 800 7 800	56 31
Coil cover	2	28			10
Strongback outside coils	28		95% JK2LB steel structure 5% liquid helium coolant	7 800	28.9
Intercoil structure	16 to 28				28.9
Cryostat	5	100	100% Type 304 stainless steel	7800	38.9

^aReplaced components.

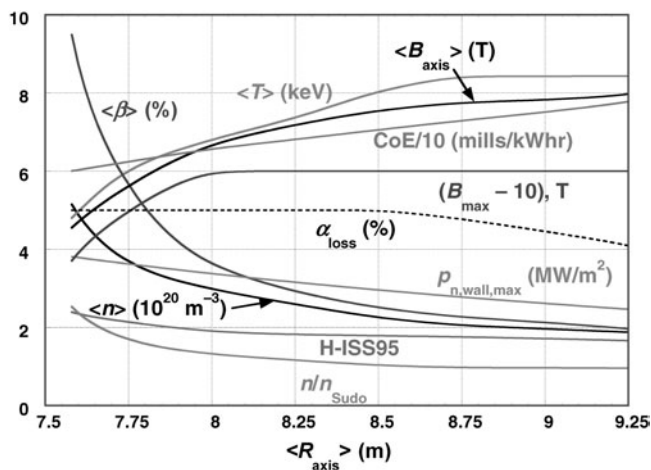


Fig. 33. Parameter variation with major radius for the reference plasma and coil configuration and the LiPb/SiC blanket and shield design.

examined has a CoE that is only 89% of the CoE for the $\langle R_{axis} \rangle = 7.75$ -m LiPb/FS/He reference case.

Only a slightly higher CoE is obtained for a full-thickness blanket and shield concept, versus a tapered blanket and shield concept, for the LiPb/SiC case. Unlike the LiPb/FS/He case, where the nominal thickness at the right-hand side of Fig. 5 is 48 cm thicker than the minimum thickness at the left-hand side, the nominal thickness at the right-hand side of Fig. 32 for the LiPb/SiC case is only 6 cm thicker than the minimum thickness at the left-hand side. The thickness for the full-blanket case is 1 cm thinner since the blanket is 4 cm thinner and the shield is 3 cm thicker, as in the LiPb/FS/He case. A further constraint was needed on $\langle B_{axis} \rangle$ to avoid too high a value for $\langle \beta \rangle$; $\langle B_{axis} \rangle$ was chosen to hold $\langle \beta \rangle$ close to that for the reference case. The value for $\langle R_{axis} \rangle$ is larger (7.9 versus 7.75 m) for approximately the same value for $\langle \beta \rangle$ for the full blanket and shield LiPb/SiC case. The full-blanket case has slightly higher

component costs and a CoE 0.3 mill/kW(electric)·h higher than the tapered blanket/shield case, as indicated in Table VII.

X.F. Other Quasi-Axisymmetric Compact Stellarator Configurations

Two other quasi-axisymmetric compact stellarator configurations were developed in the ARIES-CS study: a 24-coil $M = 3$ SNS plasma configuration and a 16-coil $M = 2$ MHH2 plasma configuration.⁷ Figure 34 shows top views of the plasma and nonplanar modular coils for these configurations and Table I compares the main dimensionless parameters for these configurations with the reference NCSX ARE configuration. The A_Δ values and whether there is room for a tapered blanket and shield determine the value for $\langle R_{axis} \rangle_{\min}$ and hence to a large extent the cost of the fusion core and the CoE. The other factor that has a significant impact on the cost of the fusion power core and hence the CoE is the cost of the coils and associated support structure through the $j_{SC}(B_{\max})$ and $\text{cost}/\text{kA} \cdot \text{m}(B_{\max})$ relations shown in Fig. 8 and the $B_{\max}/\langle B_{axis} \rangle$ curve shown in Fig. 3; other differences are less important.

The SNS and MHH2 configurations have not been developed in as much detail as the ARE configuration, so some assumptions were made to roughly model the differences. To simulate the effect of the lower α -particle losses, the same $\langle n \rangle \langle R_{axis} \rangle / \langle T \rangle^2$ dependence was assumed and the loss rate was scaled by $(\langle R_{axis} \rangle \langle B_{axis} \rangle \iota / A_p)^2$ and the base loss rates in Table I. The distributions on the wall of the radiated power and the neutron flux depend on the shape of the wall (and hence the plasma surface) and have not been calculated for these configurations. Lacking other information, the peak-to-average ratio of power densities on the wall is scaled from that

from the ARE configuration by the plasma aspect ratio and an effective multiplier K/K_{ARE} on the plasma surface area, where $K = (\text{plasma surface area})/[\text{circularized plasma surface area} = (2\pi \langle R_{axis} \rangle)^2 / A_p]$. The effect is small; the normalization is 0.921 for MHH2 and 1.007 for SNS. The value for $\langle R_{axis} \rangle$ is more important in the limiting wall power density than these differences.

The SNS configuration described in Ref. 7 is similar to the reference ARE plasma configuration but has a higher plasma aspect ratio, a flatter ι profile, and lower α -particle losses for the same collisionality. The nominal minimum distance needed for the components between the plasma edge and the center of the coil pack at $\Delta_{\min}$ for the reference tapered LiPb/FS/He blanket and shield design is 1.305 m; since $A_\Delta = 6.02$, this would give $\langle R_{axis} \rangle_{\min} = 7.86$ m versus $\langle R_{axis} \rangle_{\min} = 7.75$ m for the reference NCSX ARE configuration. However, the smaller normalized wall area in Table I for SNS would give a peak neutron wall power density of 6.9 MW/m², and the rough model discussed in the previous paragraph would further increase $p_{n,\text{wall},\text{max}}$ beyond the constrained limit. In addition, $B_{\max}$ also reaches the 16-T limit, which also indirectly constrains $\langle R_{axis} \rangle_{\min}$ by requiring a thicker coil winding pack. The value of $p_{n,\text{wall},\text{max}}$ reaches the 5.41 MW/m² limit at $\langle R_{axis} \rangle = 8.96$ m. Although the plasma radius and volume is smaller than for the reference case, the wall area is larger, which leads to higher component costs and a CoE that is 5.1 mills/kW(electric)·h higher. The other parameters for the SNS tapered LiPb/FS/He blanket and shield design are compared with the reference ARIES-CS case in Table IX. The fraction of α -particle energy lost is smaller than for the other cases.

The two-field-period MHH2 configuration described in Ref. 7 has a much lower plasma aspect ratio, simpler coils, and lower α -particle losses than the reference ARE

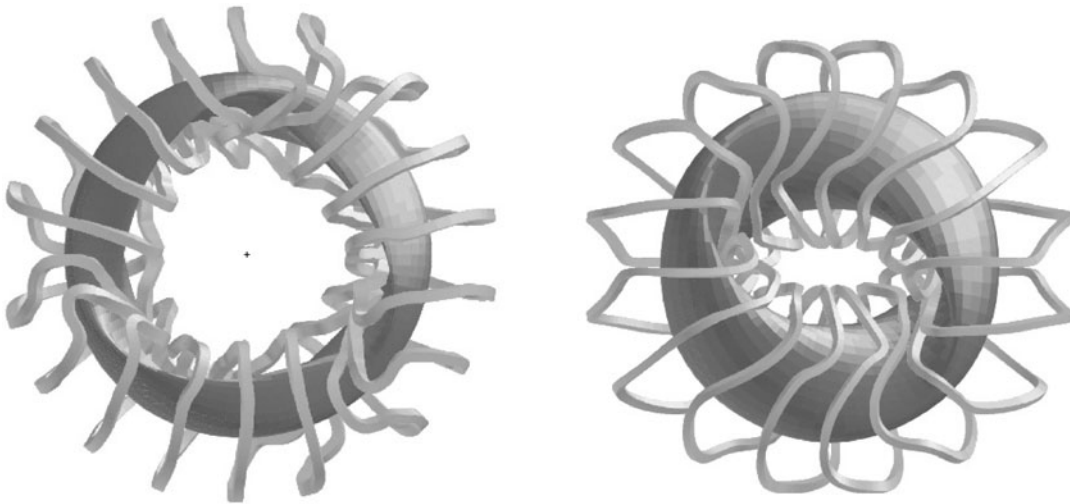


Fig. 34. Top views of the plasma and coil configurations for SNS (left) and MHH2 (right).

TABLE IX
Comparison of Plasma and Coil Configurations
for 1-GW(electric) CS Power Plants

	Reference	SNS	MHH2
Power plant parameters			
Blanket/shield type	Tapered	Tapered	Full thickness
Fusion power (MW)	2436	2436	2335
Gross electric power (MW)	1253	1253	1200
$p_{n,max}$ (MW/m ²)	5.41 ^a	5.41 ^a	2.10
Thermal efficiency (%)	43.0	43.0	43.6
CoE [mills/kW(electric)·h]	77.6	82.7	86.8
Device parameters			
$\langle R_{axis} \rangle$ (m)	7.75	8.96	9.25
$\langle a \rangle$ (m)	1.70	1.49	3.48
Wall area (m ²)	728	746	1611
$\langle B_{axis} \rangle$ (T)	5.70	6.80	3.76
B_{max} (T)	15.1	16 ^a	11.4
Plasma parameters			
$\langle n \rangle$ (10 ²⁰ m ⁻³)	4.01	3.24	1.76
$\tilde{n}/n_{Sudo}$	1.50	1.03	1.85
$\langle T \rangle$ (keV)	6.55	8.41	6.54
Plasma stored energy (MJ)	551	505	1204
Plasma volume (m ³)	444	394	2210
$\langle \beta \rangle$ (%)	6.40	4.64	6.47
H-ISS95	2.04	2.19	2.03
$f_{\alpha,loss}$ (%)	5	2.15	5
τ_E (s)	1.19	1.06	2.72
Component costs (M\$)			
Blankets and divertor	59.3	68.6	129
Shield, back wall, manifold	229	251	397
Modular coils	116	137	53.6
Coil structure	93.5	188	113
Total core cost	869	1047	1165
Heat transport	475	475	460
Reactor plant	1539	1725	1826
LiPb breeder/coolant	151	175	316
Total direct cost	2620	2829	3050
Replaced components	966	984	612

^aConstraint limit.

configuration. However, the coils are closer to the plasma on the outboard side and have less toroidal excursion than the ARE coils. Thus, the region where $\Delta < \Delta_{min} + 0.2$ m (where a tapered blanket and shield are possible) occurs over a larger area and the full-thickness blanket and shield is required everywhere. The nominal distance needed for the components between the plasma edge and the center of the coil pack for the full blanket and shield is 1.706 m; since $A_\Delta = 5.55$, this gives $\langle R_{axis} \rangle_{min} = 9.47$ m for the MHH2 configuration. The required minimum spacing can be reduced by 3.9 cm because the larger radius gives a much reduced $p_{n,wall,max}$, and hence less shield thickness is required.

The value $\langle R_{axis} \rangle = 9.25$ m satisfies the reduced space constraint, and $p_{n,wall,max}$ increases slightly to 2.1 MW/m². As with the other nontapered blanket cases, a further constraint was needed on $\langle B_{axis} \rangle$ to avoid too high a value

for $\langle \beta \rangle$; $\langle B_{axis} \rangle$ was chosen to hold $\langle \beta \rangle$ close to that for the reference case. The average plasma radius, plasma volume, stored plasma energy, and wall area are considerably larger for the MHH2 configuration due to the much smaller plasma aspect ratio and the need for a full-thickness, rather than tapered, blanket. While the much larger wall area leads to higher components costs, the low B_{max} partially offsets this by reducing the cost of the winding pack, and the low $\langle B_{axis} \rangle$ helps reduce the cost of the coil structure. The resulting CoE is 9.2 mills/kW(electric)·h more than that for the reference case, partly due to the lower cost of periodically replacing the first wall, divertor, blankets, and back wall; fewer replacements are needed because of the low neutron wall power density, resulting in a lower total cost for the replaced components. Table IX gives the other device and plasma parameters for the MHH2 configuration.

X.G. Variation with Generated Power Level

The reference case and the different blanket and shield types, as well as the different plasma and coil configurations, discussed in Secs. X.D, X.E, and X.F were for 1-GW net electric power plants. Many of the cost components scale more slowly than the power generated, so the CoE can decrease with $P_{electric}$, especially if $\langle R_{axis} \rangle$ is not constrained by the $p_{n,wall,max}$ limit. Tables VII and IX show that this is the situation for the reference ARE plasma and coil configuration for the LiPb/SiC blanket and shield design and for the ARE and MHH2 plasma and coil configurations with the LiPb/FS/He full-thickness blanket and shield. However, this is not the case for the reference ARIES-CS, where $p_{n,wall,max} = 5.41$ MW/m² at $P_{electric} = 1$ GW. Table X shows the effect of varying $P_{electric}$ from 1 to 2 GW for the reference case. The CoE [in year 2004 mills/kW(electric)·h] decreases only from 77.6 at 1 GW to 76.1 at 1.5 GW and increases slightly to 83.1 at 2 GW because of a 10-T constraint on $\langle B_{axis} \rangle$. On the other hand, for the LiPb/SiC blanket and shield case, the CoE decreases from 60.2 mills/kW(electric)·h at $P_{electric} = 1$ GW to 53.9 mills/kW(electric)·h at 1.5 GW and is approximately constant for higher power because $p_{n,wall,max}$ reaches 5.4 MW/m² at 1.5 GW. In both cases $\langle R_{axis} \rangle$ and $\langle B_{axis} \rangle$ increase and $\langle \beta \rangle$ decreases with increasing $P_{electric}$.

XI. COMPARISON WITH OTHER POWER PLANT STUDIES

The relatively low plasma aspect ratio of compact stellarators allows fusion reactors with major radii smaller than those for other stellarator reactor configurations and is comparable to those for tokamak reactors. Figure 35 illustrates the variation of $\langle R_{axis} \rangle$ with plasma aspect ratio for different types of fusion power plants. The HSR-4 and HSR-5 are modular stellarator reactors based on the W-7X configuration with $\langle R_{axis} \rangle = 18$ and 22 m, respectively.² The MHR-S and the FFHR-1 are reactors with

TABLE X
Parameter Dependence on $P_{electric}$ for the Reference Case

$P_{electric}$ (GW)	$\langle R_{axis} \rangle$ (m)	$\langle B_{axis} \rangle$ (T)	$\langle \beta \rangle$ (%)	$P_{n,wall,max}$ (MW/m ²)	CoE [mills/kW(electric)·h]
LiPb/FS/He Blanket/Shield					
1.00	7.75	5.70	6.40	5.41 ^a	77.6
1.25	8.55	7.61	3.32	5.40 ^a	77.0
1.50	9.32	8.07	2.81	5.35	76.1
1.75	10.11	10.00 ^a	1.73	5.23	81.5
2.00	10.90	10.00 ^a	1.64	5.08	83.1
LiPb/SiC Blanket/Shield					
1.00	7.75	5.64	5.45	3.62	60.2
1.25	7.75	5.64	6.14	4.70	54.8
1.50	8.10	6.89	4.19	5.40 ^a	53.9
1.75	8.74	7.75	3.15	5.37	53.6
2.00	9.42	8.27	2.63	5.21	53.9

^aConstraint limit.

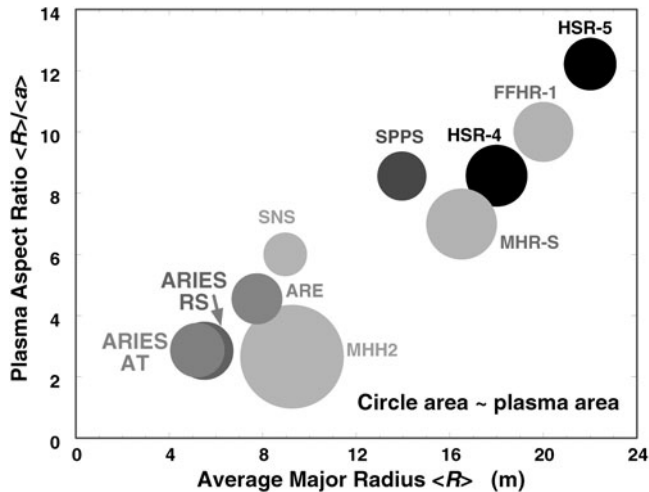


Fig. 35. Aspect ratio variation with $\langle R_{axis} \rangle$ for different reactors.

continuous helical coils based on the LHD configuration.²⁶ The reference (ARE) ARIES-CS case has the smallest $\langle R_{axis} \rangle$ and CoE of the quasi-axisymmetric compact stellarator reactors studied. The two-field-period MHH2 cannot take advantage of its lower aspect ratio because it needs a full-thickness blanket, which drives it to larger $\langle R_{axis} \rangle$ and $\langle a \rangle$, which in turn increases component costs through the larger surface area.

Tables XI and XII compare the masses and costs of the main components for the reference ARIES-CS with those for the ARIES-AT (Ref. 25) and the ARIES-RS (Ref. 17) tokamak power plants and the SPPS modular

stellarator power plant.³ In these tables the reference case (with the LiPb/FS/He blanket/shield design) is compared with ARIES-RS and SPPS, which had similar types of blanket and shields, and the ARE case with the LiPb/SiC blanket/shield design is compared with ARIES-AT, which had a similar type of blanket and shield. The ARIES-CS costs have been transformed to year 1992 dollars for comparison with the other studies using the official U.S. government inflation index. The reference ARIES-CS has parameters between those of ARIES-RS and SPPS, with most parameters closer to those for ARIES-RS. The CoE for the reference ARIES-CS is less because of the higher plant availability and lower cost for replaced components. Costs of fabricated components have been updated to year 2004 material costs, and improvements were made in the blanket and shield concept.¹¹ Also, an advanced approach²³ was assumed for fabricating the massive coil support structure; a more conventional approach would increase the CoE by 2 mills/kW(electric)·h. The ARIES-AT tokamak power plant has more differences with an ARE case that has a LiPb/SiC blanket and shield. The ARE case has a considerably higher net efficiency and lower component replacement costs. Both ARIES-CS cases, the reference case with the LiPb/FS/He blanket/shield and that with the advanced LiPb/SiC blanket/shield, are competitive with their tokamak power plant counterparts.

XII. SUMMARY AND CONCLUSIONS

A stellarator systems/optimization code was used to optimize the ARIES-CS fusion power plant parameters

TABLE XI
Mass Comparisons with Previous ARIES Power Plant Studies

	ARIES-CS	ARIES-RS	SPPS	ARIES-CS	ARIES-AT
Configuration	Compact stellarator	Tokamak	Modular stellarator	Compact stellarator	Tokamak
Blanket/shield case	LiPb/FS/He			LiPb/SiC	
Average major radius $\langle R_{axis} \rangle$ (m)	7.75	5.52	13.95	7.75	5.20
Plasma volume (m ³)	444	349	734	444	329
Wall area (m ²)	728	436	1 171	728	426
Average field on-axis $\langle B_{axis} \rangle$ (T)	5.70	7.98	4.94	5.70	5.86
Volume-averaged $\langle \beta \rangle$ (%)	6.4	5.0	5.0	6.4	9.2
Mass (tonnes)					
First wall/blanket/back wall	663	585	251	614	255
Shield, back wall, manifold	4 585	4 235	9 453	2 619	882
Coils	627	3 280	4 191	614	1 525
Coil structure	3 465	1 627	5 365	3 390	—
Vacuum vessel	1 440	1 357	2 171	1 198	1 415
Fusion power core	13 688	12 679	21 430	9 990	5 226

TABLE XII
Cost Comparisons with Previous ARIES Power Plant Studies in 1992 Dollars

Parameter	ARIES-CS	ARIES-RS	SPPS	ARIES-CS	ARIES-AT
Blanket/shield case	LiPb/FS/He			LiPb/SiC	
Fusion power core mass (tonnes)	13 688	12 679	21 430	9 990	5 226
Thermal efficiency	43%	46%	46%	60%	59%
Plant availability	85%	76%	76%	85%	85%
Net efficiency	34.3%	38.2%	43.6%	56.7%	50.4%
CoE [mills/kW(electric)·h; 1992 \$]	61.3	75.8	74.9	48.0	47.5
90 Total direct cost (M\$)	2 071	2 194	2 261	1 485	1 521
20 Land	10.2	10.4	10.4	10.2	10.6
21 Structure	265	331	333	214	254
22 Reactor plant equipment	1 218	1 390	1 487	823	761
23 Turbine plant equipment	248	284	254	215	243
24 Electric plant equipment	110	111	104	89.6	98.5
25 Miscellaneous plant equipment	56.0	56.2	51.9	47.7	47.4
26 Special materials	119	11.1	21.1	66.1	83.8
27 Heat rejection	44.3	—	—	18.9	23.3

for minimum CoE and $\langle \beta \rangle$ subject to a large number of physics, engineering, and in-vessel component constraints. The most important factors determining the size of the reference case are the allowable $p_{n,wall,max}$, the distance needed between the edge of the plasma and the nonplanar magnetic field coils for the intervening components and regions, and an adequate TBR. The magnetic field and coil parameters are determined by both plasma performance and constraints on the coil pack. Optimization (e.g., minimization of the CoE) is not just a matter of low plasma aspect ratio. Even more important is having a low plasma-coil distance aspect ratio and a

coil geometry that allows an adequate tapered blanket/shield region. Although the MHH2 plasma and coil configuration has a plasma aspect ratio 58% of that for the ARE configuration and a plasma-coil distance aspect ratio 7% smaller, the inability to have a tapered blanket/shield design drives the size and CoE for an MHH2 power plant higher than that for even an SNS-based power plant, where the plasma aspect ratio is a factor of 2.26 higher and the plasma-coil distance aspect ratio is 8.5% higher.

The same costing approach and algorithms used in previous ARIES studies with updated material costs lead to a reference compact stellarator power plant with

$\langle R_{axis} \rangle = 7.75$ m, $\langle B_{axis} \rangle = 5.7$ T, $\langle \beta \rangle = 6.4\%$, and a projected CoE = 77.6 mills/kW(electric)·h with a LiPb/FS/He blanket and shield design. A number of physics assumptions (density and temperature profiles, impurity radiation levels, confinement enhancement, $\langle \beta \rangle$ limits, alpha-particle losses, etc.) affect the plasma performance but not the size and cost of the fusion power core. Other parameters (B_{max} at the coils, component cost penalties, different blanket and shield approaches, alternative plasma and coil configurations, etc.) have a larger impact. In particular, use of an advanced LiPb/SiC blanket and shield concept like that adopted in the ARIES-AT tokamak power plant study could reduce the CoE by ~ 17 mills/kW(electric)·h.

XII.A. Open Issues for Compact Stellarator Research from the Systems Studies

The ARIES-CS systems studies raise a number of issues for development of compact quasi-axisymmetric stellarators.

1. Can $\langle \beta \rangle \sim 6\%$ be achieved and sustained with good confinement? What is the maximum useful $\langle \beta \rangle$ value?
2. Can a low fast-ion (alpha-particle) loss rate be achieved? Can fast-ion losses due to MHD instabilities be mitigated by operation at high density?
3. Can a workable divertor configuration with moderate size and power peaking be developed that controls impurities and enables pumping of the neutral particles and helium ash?
4. Can regimes of minimal power excursions onto the first wall (e.g., due to disruptions and edge-localized modes) be achieved?
5. Under what conditions can acceptable plasma purity and low impurity accumulation be achieved?
6. Is the energy confinement at least 1.5 times ISS95 scaling? How does it extrapolate to larger size?
7. What are other operational limits (density, controllable core radiation fraction, etc.)? Can a high radiation fraction be sustained in the plasma? (From the edge?) Can a high fraction of power in the SOL be radiated to the wall and avoid a high thermal power flux on the divertor?
8. How do the density and pressure profile shapes depend on configuration and plasma parameters?
9. Can the coil designs be simplified? Can physics requirements be relaxed, by reduction of external transform, elimination of stability from optimization, reduction of flux surface quality requirements, or increase of helical ripple?
10. What plasma control elements and diagnostics are required?

Finally, only quasi-axisymmetric compact stellarator configurations were examined in this study. Do other quasi-symmetric (quasi-poloidal or quasi-helical) compact stellarator configurations offer a different approach to an attractive compact stellarator fusion power plant?

ACKNOWLEDGMENTS

This research was supported by the U.S. Department of Energy under contract DE-AC05-00OR22725 with UT-Battelle, LLC. The authors have benefited from discussions with other members of the ARIES-CS group, particularly F. Najmabadi, R. Raffray, T.-K. Mau, and X. Wang, and with R. Miller (Decsys Systems, Santa Fe, New Mexico).

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